

Scheduling in Broadcast Networks

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Abstract: Broadcasting in a communications network has been the subject of many studies in recent years. The studies vary in their assumptions governing the behavior of the network and in their objectives with respect to the network. Almost all the work to date uses the unit transmission time assumption, that is, the message transmission times between all pairs of vertices are equal. In this paper, we investigated the broadcast problem under four more general transmission time assumptions. In addition, four different objective functions were considered, including the minimization of (1) the broadcast time (the maximum time for any vertex to receive the message), (2) the average time to receive the message (both with and without ready times at the vertices), (3) the weighted average time to receive the message, and (4) the cycle time. Of the 20 problems thus generated, four admit polynomial time algorithms, 15 are (in the equivalent recognition version) unary *NP*-complete, and the complexity status of one remains open. All these results are proved, and heuristics with an attainable constant worst-case performance ratio are provided for two of the problems for which polynomial time algorithms are not found. © 1998 John Wiley & Sons, Inc. *Networks* 32: 233–253, 1998

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1. INTRODUCTION

A major focus in the information technology field is the design and management of computer networks, as suggested by the following quotation: “Daily, huge quantities of information are moved between computers over communication channels. . . . PC users can readily access mainframes and capture information as diverse as stock market quotes, product prices, and sports results.

PC’s are . . . tied together through networks . . .” (Hicks [6]).

From both the hardware and software standpoints, there has been enormous progress in the field of communication networks. Yet, as we discuss in our literature review, there are large, virtually unresearched areas in the design of broadcast systems which are awaiting investigation. Here, we discuss one such research area.

Specifically, the problem of constructing a broadcast schedule in a distributed system consisting of $n + 1$ processors is considered. In such a network, a message must be broadcast from one of the processors (0, the *initiator*) to all the other processors 1, . . . , n . To broadcast, it is necessary for the initiator to begin by transmitting the message to processors adjacent to it in the network. These

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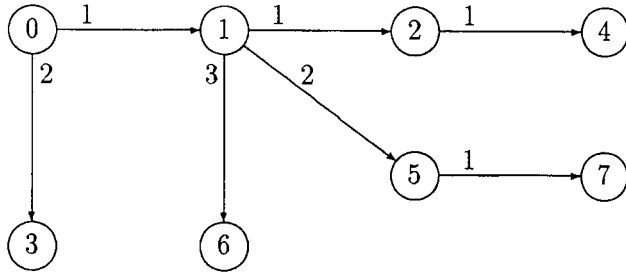


Fig. 1. An example broadcast network.

processors may, in turn, transmit the message to other processors, until all processors have received the message. The problems considered here deal with systems in which a direct transmission from (processor) i to (processor) j (i.e., a transmission which goes from i to j with no intermediate processors) occupies both processors for the full length of the transmission time. Thus, neither processor can transmit or receive any other message during the transmission from i to j . In the broadcasting literature, this is termed *processor-bound* transmission. At the completion of the transmission, both processors are free to transmit or receive other messages. We assume that each vertex can send a message directly to each other vertex.

In the model under study, processor 0 is assumed to initiate the broadcasting process at time 0, at which time processor 0 has a message which must be transmitted to all other processors. Let b_i be the *reception time* for processor i , that is, the time at which the transmission of the message to processor i is completed. The broadcast problem is to schedule the direct transmissions in order to minimize a real-valued objective function $f(b_1, \dots, b_n)$ of the vector of reception times. We shall assume, unless otherwise noted, that it is possible to make a direct transmission from i to j , for $i \neq j$, $0 \leq i \leq n$ and $1 \leq j \leq n$.

Example 1. Suppose that there are seven processors 1, $\dots$, 7 plus the initiator and that the direct transmission time from processor i to j is U_j . Figure 1 gives a representation of a not necessarily optimal broadcasting scheme. The vertex numbers represent processors, and the arc numbers give the order of transmission from each vertex. Processor 0 first transmits to processor 1. Both processors become free at time $b_1 = U_1$. At time U_1 , processor 0 immediately begins transmission to processor 3, while processor 1 begins transmission to processor 2. Processors 0 and 3 become free at time $b_3 = U_1 + U_3$, and processors 1 and 2 become free at time $b_2 = U_1 + U_2$. Processor 1 next transmits to processor 5, and processor 2 transmits to processor 4. Finally, processor 1 transmits to processor 6 while processor 5 transmits to processor 7. It is possible to compute the broadcast time vector: $b = (b_1, \dots, b_7)$

$$= (U_1, U_1 + U_2, U_1 + U_3, U_1 + U_2 + U_4, U_1 + U_2 + U_5, U_1 + U_2 + U_5 + U_6, U_1 + U_2 + U_5 + U_7).$$

We shall adopt the language and notation of graph theory in our discussions. A processor will also be called a vertex. As illustrated in Example 1, a *feasible solution* $\Pi = (\mathcal{T}, \gamma)$ for a problem defined on vertices (processors) $0, \dots, n$ will be described by

- (i) A *broadcast tree* $\mathcal{T} = (T, A(T))$ rooted at vertex 0. The tree is a directed tree with vertex set $T = \{0, \dots, n\}$ and edge set $A(T)$. Thus, $(i, j) \in A(T)$ means that i transmits the message directly to j . “Transitive edges” are not included in the edge set. Vertex i is a direct predecessor (respectively, direct successor) of j if (i, j) [respectively, (j, i)] $\in A(T)$. Predecessor, successor, and other graph theory terms are defined in the usual manner.
- (ii) For any vertex i which has successors, a specification of the *order* in which i is to transmit to its successors. Let γ denote the set of specifications (γ will simply give a permutation of successors for each vertex i which has successors). More specifically, $\gamma = (\gamma_0, \dots, \gamma_n)$ where γ_i is a permutation of the immediate successors of i in $\mathcal{T}$.

In Theorem 1 and Corollary 1, we will justify this definition of feasible solution by showing that only tree solutions without unnecessary delays between transmissions need be considered.

As an example, the solution illustrated in Figure 1 may be specified as follows: $\Pi = (\mathcal{T}, \gamma)$, where $T = \{0, 1, 2, 3, 4, 5, 6, 7\}$, $A(T) = \{(0, 1), (0, 3), (1, 2), (1, 5), (1, 6), (2, 4), (5, 7)\}$, $\gamma = (\gamma_0, \dots, \gamma_7)$, $\gamma_0 = (1, 3)$, $\gamma_1 = (2, 5, 6)$, $\gamma_2 = (4)$, $\gamma_5 = (7)$, $\gamma_3 = \gamma_4 = \gamma_6 = \gamma_7 = \emptyset$. Ultimately, any algorithm that finds a feasible solution will construct $\mathcal{T}$ and γ .

For $i \in T$, let $s(i)$ denote the set of direct successors of i , that is, $j \in s(i)$ whenever $(i, j) \in A(T)$. Let $\sigma(i) = |s(i)|$ denote the cardinality of $s(i)$. For $i \in T$, the subtree $\mathcal{T}(i)$ is defined by the vertex set $\{j \mid j = i \text{ or } j \text{ is a successor (not necessarily direct) of } i\}$ and its associated edges. The phrase i transmits to $\mathcal{T}(j)$ is interpreted to mean that i is a direct predecessor of j or, equivalently, that i transmits directly to j . In Figure 1, $s(1) = \{2, 5, 6\}$, $\sigma(1) = 3$, and $\mathcal{T}(1) = \{1, 2, 4, 5, 6, 7\}$.

There has been a significant amount of research activity on broadcasting in communications networks, especially over the last decade. In their excellent survey paper which cites 135 sources, Hedetniemi et al. [5] indicated that almost all of the literature focuses upon models subject to four common constraints:

- (a) Each transmission involves only two vertices;
- (b) Each transmission requires one unit of time;

- (c) A vertex can participate in only one transmission per unit of time;
- (d) A vertex can only transmit to an adjacent vertex (this is called *local broadcasting*).

We continue to summarize some relevant discussion from [5]. The minimum number of time units required to complete broadcasting when vertex v is the initiator is denoted by $b(v)$. It is obvious that, for any graph, $b(v) \geq \lceil \log_2 n \rceil$. The broadcast time of a graph G , denoted by $b(G)$, is equal to the maximum broadcast time of any vertex v in G , that is, $b(G) = \max \{b(v) : v \text{ is a vertex in } G\}$. A *minimal broadcast graph* is a graph G with n vertices such that $b(G) = \lceil \log_2 n \rceil$, but for any proper spanning subgraph $G' \subset G$, $b(G') > \lceil \log_2 n \rceil$. The *broadcast function* $B(n)$ is the minimum number of edges in a minimal broadcast graph having n vertices.

Some of the most important problems dealt with in the literature are the following:

- (a) Find the value of $b(v)$ for a vertex v in a graph $G = (N, A)$.
- (b) Find the broadcast center of a graph, that is, the set of vertices for which $b(v)$ is minimal.
- (c) Find the set of minimal broadcast trees in an n vertex graph, where a broadcast tree is minimal if there is a vertex i such that $b(i) = \lceil \log_2 n \rceil$.
- (d) Find $B(n)$ and the graphs which achieve this value.
- (e) Design heuristics to find minimal broadcast graphs with close to $B(n)$ edges.

Multiple message broadcasting involves information which is too extensive to be transmitted in one message. Key problems dealt with in this area involve finding:

- (a) The maximum number of vertices which can receive m messages in t time units;
- (b) The maximum number of messages which can be broadcast to p vertices in t time units;
- (c) The minimum number of time units necessary to broadcast m messages to p vertices.

A result related to our work in the area of computational complexity was obtained by Slater et al. [12], who proved the *NP*-completeness of the following problem: Given an arbitrary graph $G = (N, A)$, a vertex v , and an integer $k \geq 4$, decide whether $b(v) \geq k$.

Three recent papers discussed broadcasting with non-unit transmission time functions. Farley and Proskurowski [1] recognized different components of the time to transmit messages: switching delays at vertices and sending delays over transmission lines. However, by contrast with our work, all vertices experience the same delay function.

Koh and Tcha [7] found the broadcast center of a tree and the associated value $b(v)$ for any vertex in the broadcast center, with arbitrary times t_{ij} for direct transmissions. Their algorithm takes $O(n \log n)$ time. In contrast with this paper, their broadcast tree is given in advance. Fraignaud and Lazard [2] discussed a variety of problems with transmission times which depend linearly on the size of a message but are independent of the sending and receiving vertices. Since we do not consider messages of varying size, the unit and linear transmission time functions are identical in our context.

To summarize, the existing literature on broadcasting with nonunit transmission times is rather sparse, and the research we present extends existing results to more general transmission models. The optimal solution to a broadcast problem depends heavily upon the objective function and the direct transmission time between each pair of processors i and j . In Section 2, the set of problems under consideration is defined in terms of their objective functions and transmission time functions. Applications of the models under study are discussed, and the algorithms and complexity results of the paper are announced. In Section 3, optimal polynomial time algorithms will be presented. Section 4 gives the proofs of the complexity results announced in Section 2. In Section 5, we describe heuristics for two of the problems for which polynomial algorithms are not known and analyze the worst-case performance of those heuristics. Finally, Section 6 contains summary remarks and directions for further study.

2. OVERVIEW OF THE RESULTS

Here, we summarize our results about the complexity of broadcasting with general transmission times. As previously discussed, vertex 0 is the initiator of the message. Let U_j , $0 \leq j \leq n$, denote a parameter associated with vertex j (alternatively, processor j) which may affect transmission times into and out of that vertex. We consider four possible assumptions about the time t_{ij} required for vertex i to transmit the message to vertex j . These assumptions are, for $0 \leq i \neq j \leq n$:

- $t_{ij} = U_i$, where transmission time depends only upon the sending vertex.
- $t_{ij} = U_j$, where transmission time depends only upon the receiving vertex.
- $t_{ij} = \max \{U_i, U_j\}$, where time depends upon the slower of the two vertices.
- $t_{ij} = f(U_i, U_j)$, where f is an arbitrary, positive-valued function.

Without loss of generality, unless otherwise specified, it is assumed throughout the paper that $U_1 \leq \dots \leq U_n$,

where the position in which U_0 would appear in this sequence is arbitrary.

The basic flow model above, with a primary initiator which broadcasts through a tree to many sites, characterizes various news groups' broadcasting systems (e.g., Unix mail group). The time model $t_{ij} = U_i$ would reflect systems with large receiver buffers (adequate to hold entire messages) and fast receiver capability to process the contents of those buffers. The $t_{ij} = U_j$ model may characterize a rendezvous system in which the receiver must return to the sender a confirmation message (handshake) upon reception of the message. The $t_{ij} = \max\{U_i, U_j\}$ model would occur in a flow-controlled system, in which the slower of the two processors sets the speed. The $t_{ij} = f(U_i, U_j)$ model is a general one, covering all other eventualities.

Under each of these assumptions, we consider four objective functions familiar from the literature of machine scheduling. See Lawler et al. [10] for a thorough survey. The standard classification scheme described by Lawler et al. [9] for such problems can be denoted $\alpha|\beta|\gamma$, where α indicates the scheduling environment, β describes the job characteristics or restrictive requirements, and γ describes the objective function to be minimized. We use B to denote the broadcasting environment. The relevant assumption about transmission times t_{ij} will be described under β . The data for the problems that we consider are specified by t_{ij} and by an objective function.

Before specifying the objective functions, we define the following parameters for a problem instance:

- w_j = a relative value (a positive-valued weight) for vertex j .
- r_j = a ready time, at which vertex j first becomes available to receive the message.

We also define the following variable:

- b_j = the time at which vertex j finishes receiving the message.

We define four different objective functions:

$\sum_{1 \leq j \leq n} b_j$ = the total of the reception times to the vertices, or *total broadcast time*; minimizing this measure is also equivalent to minimizing the mean reception time to the vertices.

$\sum_{1 \leq j \leq n} w_j b_j$ = the total of the weighted reception times to the vertices, or *total weighted broadcast time*, where w_j is a positive number.

$\max_{1 \leq j \leq n} \{b_j\}$ = the maximum reception time, which represents the elapsed time from start to completion of the broadcast, or *maximum broadcast time*.

b' = the broadcast cycle time, or period, of the net-

work, which represents the minimum time between the broadcast of two consecutive messages in the network.

The fourth objective function, b' , requires some explanation: Given a feasible solution, suppose that we wish to transmit several messages, one after the other, each broadcast to be implemented according to some feasible solution. An obvious lower bound on the amount of time between successive broadcasts is the sum of the transmission times out of vertex 0. More generally, let i be the direct predecessor of j , and recall that $s(j)$ is the set of direct successors to j . Then, j is "busy" for $c_j = t_{ij} + \sum_{k \in s(j)} t_{jk}$ time units. Thus, two successive broadcasts cannot occur closer in time than $\max_{0 \leq j \leq n} \{c_j\}$ without interference. In Corollary 1 at the end of this section, we show that these c_j time units may be assumed without loss of generality to take place without inserted idle time, that is, the busy period for vertex j in each broadcast extends over exactly c_j time units. Consequently, the lower bound on b' is attainable by a feasible solution, that is, $b' = \max_{0 \leq j \leq n} \{c_j\}$. Thus, our fourth objective function is

b' = the maximum busy period of any vertex, or *cycle time*.

Before proceeding further, we provide an example to illustrate how the two transmission models $t_{ij} = U_i$ and $t_{ij} = U_j$ may yield different optimal solutions even with the same objective function. Consider the problem instance given by $n = 3$, $(U_0, U_1, U_2, U_3) = (10, 1, 10, 10)$, with the objective to minimize $\sum_{1 \leq j \leq n} b_j$. Under the $t_{ij} = U_i$ model, an optimal solution is for 0 to transmit to 1 and for 1 to transmit to 2 and then to 3. This solution may be written as $\Pi = (\mathcal{I}, \gamma)$, where $T = \{0, 1, 2, 3\}$, $A(T) = \{(0, 1), (1, 2), (1, 3)\}$ and $\gamma = (\gamma_0, \gamma_1, \gamma_2, \gamma_3) = ((1), (2, 3), \emptyset, \emptyset)$. The reception times are $(b_1, b_2, b_3) = (10, 11, 12)$. Under the $t_{ij} = U_j$ model, an optimal solution is for 0 to transmit to 1 and then to 3 and for 1 to transmit to 2. This solution may be written as $\Pi = (\mathcal{I}, \gamma)$, where $T = \{0, 1, 2, 3\}$, $A(T) = \{(0, 1), (0, 3), (1, 2)\}$, and $\gamma = (\gamma_0, \gamma_1, \gamma_2, \gamma_3) = ((1, 3), (2), \emptyset, \emptyset)$. The reception times are $(b_1, b_2, b_3) = (1, 11, 11)$.

The total broadcast time objective is considered, first with all vertices becoming available simultaneously, and, second, with nonzero ready (arrival) times, which we denote by $r_j \geq 0$ under β . Thus, for example, we use $B|t_{ij} = U_i, r_j| \sum_{j=1}^n b_j$ to denote the problem of minimizing total broadcast time, in the presence of ready times for vertices, when transmission times depend only upon the sending vertex.

The objective of minimizing the maximum broadcast time ($\max_{1 \leq j \leq n} \{b_j\}$) is of obvious importance and is the

TABLE I. Complexity of broadcasting problems

	Objective				
	Σb_j	$r_j \Sigma b_j$	$\Sigma w_j b_j$	$\max_{1 \leq j \leq n} \{b_j\}$	b^t
$t_{ij} = U_i$	$O(n \log n)$ Thm 2	U Thm 6	U Thm 8	U Thm 10	Open
$t_{ij} = U_j$	$O(n \log n)$ Thm 3, Cor 2	U Thm 7, Cor 3,	U Thm 9, Cor 5,	U Thm 11, Cor 7,	$O(n \log n)$ Thm 4
$t_{ij} = \max\{U_i, U_j\}$		Cor 4	Cor 6	Cor 8	
$t_{ij} = f(U_i, U_j)$	U Thm 5				U Thm 12, Cor 9

most widely studied of our five objectives, as discussed in the literature review in Section 1. The cycle time objective (b^t) is important to consider, because it is a useful measure of the capacity of the system to carry many messages over time (again, see Section 1). The weighted sum of broadcast times ($\sum_{i=1}^n w_j b_j$) models the situation in which the processors have varying costs of availability (w_j) per unit time. The unweighted sum of broadcast times ($\sum_{i=1}^n b_j$) models the special case where all processors have equal costs. The addition of ready times r_j may be used to model the case in which processors are committed to previous jobs and cannot start to receive the message until those jobs have completed processing.

The model under discussion may be interpreted in another way. Suppose that the sites in a communication network have been determined, along with the computer facilities located at those sites, and it is desired to connect the sites together as cheaply as possible, in order to achieve one of the broadcast objectives that we discuss. Instead of deciding which of many links to use for message broadcasts, the problem becomes the determination of which links to construct and use. Thus, the scheduling problem becomes a design and schedule problem.

Table I is a matrix in which the rows represent the four different assumptions about t_{ij} , and the columns represent our five different combinations of objective functions and assumptions about ready times. We provide the time complexity of polynomial time algorithms, and we use U to indicate that (the equivalent recognition version of) a problem is unary NP -complete (see Garey and Johnson [3] for related definitions). Of the 20 problems thus generated, four admit polynomial time algorithms, 15 are proved to be unary NP -complete, and one problem remains open. Included in the appropriate cell in Table I is a reference to a theorem in this paper that contains a proof of this result. In cases where a result implies others, the lines dividing the relevant cells in Table I have been omitted.

The only problem in Table I that remains open is $B | t_{ij} = U_i | b^t$. We conjecture that the recognition version of that problem is NP -complete, at least in the ordinary sense. The description and worst-case analysis of heuristics

for that problem, and for $B | t_{ij} = U_j | \max_{1 \leq j \leq n} \{b_j\}$, can be found in Section 5.

To summarize, a broadcast is accomplished by a set of direct transmissions, where (i) every vertex $j \neq 0$ must be on the receiving end of one such transmission; (ii) no vertex i may transmit to another vertex until it has completed reception of the message; and (iii) no vertex is simultaneously involved in more than one transmission at a time, whether as receiver or sender. A feasible solution as previously defined gives one possible way to broadcast to all vertices. For each of the 20 problems, we wish to find the optimal solution, that is, the best feasible solution.

We now present an important result which has a bearing on all the remaining sections:

Theorem 1. *There exists an optimal solution to each problem in Table I such that the direct transmission links used in the optimal solution form a broadcast tree oriented away from vertex 0.*

Proof. If a processor were to receive the message more than once in the problems specified in this table, all but the first transmission could be dropped with no increase in the objective function. ■

Consider an optimal solution $\Pi = (\mathcal{I}, \gamma)$ for any of the problems in Table I. In our discussion of the definition of b^t earlier in this section, we noted that each vertex $j \in \{0, \dots, n\}$ is busy for $c_j = t_{ij} + \sum_{k \in s(j)} t_{jk}$ time units, where i is the direct predecessor of j and where t_{ij} is defined to be 0 for $j = 0$.

Corollary 1. *There exists an optimal solution for each problem in Table I such that the busy time c_j for each vertex j extends over a period of precisely c_j time units, that is, there is no idle time for vertex j from the time i starts its broadcast to j until the time j has completed all its transmissions.*

Proof. By Theorem 1, the optimal solution is in the form of a tree oriented away from vertex 0. Therefore, each vertex $j \in \{1, \dots, n\}$ receives precisely one mes-

sage, so no message needs to be delayed because of interference at the receiving vertex. Hence, any existing idle time (as described in the statement of the corollary) may be removed, and the objective function in the resulting schedule will be no larger than in the original schedule. ■

Theorem 1 and Corollary 1 allow us to restrict our attention to feasible solutions of the form $\Pi = (\mathcal{T}, \gamma)$, where $\mathcal{T}$ is a broadcast tree and γ is a set of permutations of the direct successor sets, one permutation for each vertex.

3. POLYNOMIAL TIME ALGORITHMS

We present the three polynomial time algorithms that are indicated in Table I.

3.1. $B | t_{ij} = U_i | \sum_{j=1}^n b_j$

We describe a polynomial time “greedy” algorithm, ISP1, for the problem of minimizing the total broadcast time, where transmission times depend only upon the sending vertex. This algorithm induces transmissions to the vertices in nondecreasing order of U_i ; since we have adopted the convention that $U_1 \leq \dots \leq U_n$, this is equivalent to transmitting to the vertices in the order 1, $\dots$, n . Thus, when it is vertex i 's turn to receive the transmission, vertices 0 to $i - 1$ will already have received it. Vertex i will be assigned to receive the transmission from the vertex which will yield the smallest reception time for vertex i .

In ISP1, A_i is the last time at which vertex i was busy with either a transmission or a reception. A_i is updated whenever i either receives or transmits a message.

Algorithm Iterated Shortest Path 1 (ISP1)

Given: positive numbers $U = U_i$, $0 \leq i \leq n$.

1. Initialize: $A_0 = \dots = A_n = 0$, $A(T) = \emptyset$, $\gamma_1 = \dots = \gamma_n = \emptyset$, $b_0 = 0$.
Order the numbers such that $U_1 \leq \dots \leq U_n$.
2. For $1 \leq i \leq n$
 $b_i = \min \{A_j + U_j | 0 \leq j \leq i - 1\}$.
 $p = \min \{j | 0 \leq j \leq i - 1, b_i = A_j + U_j\}$.
(i becomes the next vertex to which p will transmit).
 $A_i = b_i$, $A_p = b_i$, $A(T) = A(T) \cup \{p, i\}$, $\gamma_p = (\gamma_p, i)$.
 EndFor.

Terminate.

Example 2. Consider the problem given by the following data: $U = (U_0, \dots, U_8) = (7, 5, 6, 7, 8, 10, 11, 13, 15)$.

The broadcast tree produced by Algorithm ISP1 is illustrated in Figure 2, where each rectangular vertex j contains the number, the transmission time U_j , and the reception time b_j , of vertex j , respectively. First, vertex 0 transmits to vertex 1, so $b_1 = 7$. Both vertices become free at time 7; hence, $A_0 = A_1 = 7$. Next, to decide whether vertex 0 or vertex 1 should transmit to vertex 2, we note that the corresponding reception times at vertex 2 would be $14 = A_0 + U_0$ from vertex 0 and $12 = A_1 + U_1$ from vertex 1. So, we choose to transmit from vertex 1; we set $b_2 = 12$ and $A_1 = A_2 = 12$. Similarly, for vertex 3, we compare $14 = A_0 + U_0$ from vertex 0, $17 = A_1 + U_1$ from vertex 1, and $18 = A_2 + U_2$ from vertex 2. So, vertex 0 transmits to vertex 3, $b_3 = 14$, and $A_0 = A_3 = 14$. Continuing thus yields the broadcast tree of Figure 2.

A *multiset* is a set which can include more than one copy of the same element. In Lemmas 1 and 2, the union of two sets is a multiset union: If $C = A \cup B$, then the multiplicity of any element $x \in C$ is the sum of its multiplicities in A and in B . The term “distinct elements,” when applied to a multiset, could refer to two identical values if the multiplicity of that value exceeds 1. For example, if $C = \{1, 2, 2, 3, 5\}$, then 2 and 2 are two distinct elements of C .

In the proofs below, notation will be adopted to “match.” For example, if four different broadcast trees B, B^1, B^* , and B^2 are under discussion, then the reception times in these trees will be denoted by b_i , b_i^1 , b_i^* , and b_i^2 , respectively. Lemmas 1 and 2, which now follow, will be used in the proof of optimality for ISP1.

Lemma 1. Suppose that ϕ_1 and ϕ_2 are positive integers, $0 < \delta < \lambda$, and that $0 < \Delta < \Lambda$. Let $\mathcal{N}_j = \{1, \dots, j\}$ be the set of natural numbers from 1 through j , where we define $\mathcal{N} = \mathcal{N}_\infty$. Let $(\beta'_1, \dots, \beta'_{\phi_1+\phi_2})$ be a permutation, in nondecreasing order, of the $\phi_1 + \phi_2$ smallest elements of the multiset $\{\delta + i\Delta | i \in \mathcal{N}\} \cup \{\lambda + i\Delta | i \in \mathcal{N}\}$. Let $(\beta_1, \dots, \beta_{\phi_1+\phi_2})$ be any permutation of any $\phi_1 + \phi_2$ distinct elements of the multiset $\{\delta + i\Delta | i \in \mathcal{N}\} \cup \{\lambda + i\Delta | i \in \mathcal{N}\}$. Then, for $1 \leq j \leq \phi_1 + \phi_2$, the inequality $\sum_{i \in \mathcal{N}_j} \beta_i \geq \sum_{i \in \mathcal{N}_j} \beta'_i$ holds.

Proof. For each j there exist nonnegative integers e and f , $e + f = j$, such that $\sum_{i \in \mathcal{N}_j} \beta_i$ is the sum of e elements from the set $\{\delta + i\Delta | i \in \mathcal{N}\}$ and f elements from the set $\{\lambda + i\Delta | i \in \mathcal{N}\}$. By choosing the smallest elements from each of these sets, we get a lower bound:

$$\begin{aligned} \sum_{i \in \mathcal{N}_j} \beta_i &\geq \sum_{i \in \mathcal{N}_e} (\delta + i\Delta) + \sum_{i \in \mathcal{N}_f} (\lambda + i\Delta) \\ &= e\delta + e(e+1)\Delta/2 + f\lambda + f(f+1)\Delta/2 \equiv LB. \end{aligned}$$

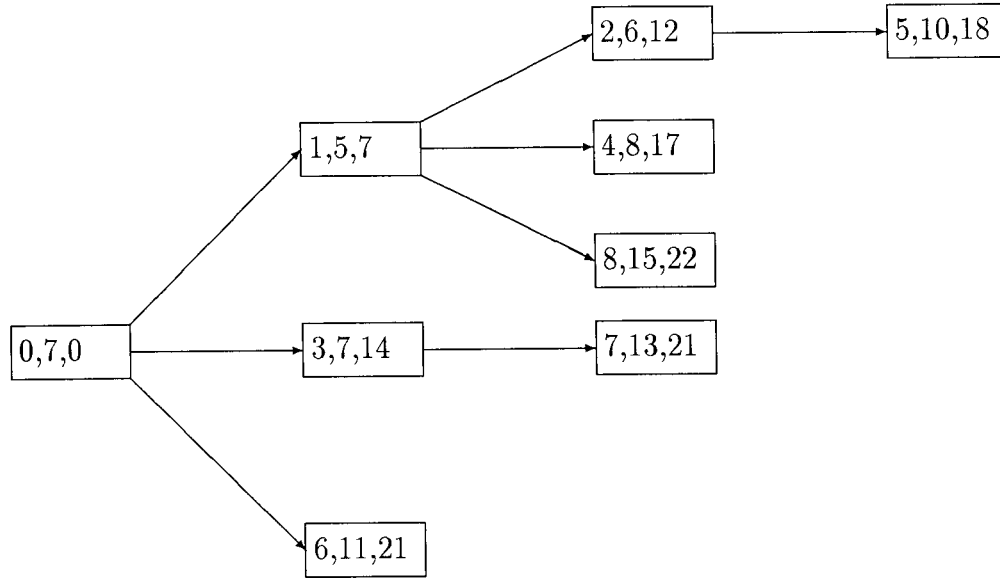


Fig. 2. Application of Algorithm ISP1 to Example 2.

On the other hand,

$$\begin{aligned}
 \sum_{i \in \mathcal{N}_j} \beta'_i &\leq \min \left\{ \sum_{i \in \mathcal{N}_e} (\delta + i\Delta) + \sum_{i \in \mathcal{N}_f} (\lambda + i\Lambda), \right. \\
 &\quad \left. \sum_{i \in \mathcal{N}_f} (\delta + i\Delta) + \sum_{i \in \mathcal{N}_e} (\lambda + i\Lambda) \right\} \\
 &= \min \{ e\delta + e(e+1)\Delta/2 + f\lambda + f(f+1)\Lambda/2, \\
 &\quad f\delta + f(f+1)\Delta/2 + e\lambda + e(e+1)\Lambda/2 \} \\
 &\equiv \min \{ UB_1, UB_2 \}.
 \end{aligned}$$

It is straightforward to verify that $LB \geq UB_1$ for $e \geq f$ and $LB \geq UB_2$ for $e \leq f$, and the lemma follows. ■

Lemma 2. Given the hypothesis of Lemma 1 and also that $t_1 \geq \dots \geq t_{\phi_1+\phi_2} > 0$, then

$$\sum_{j=1}^{\phi_1+\phi_2} \beta_j t_j \geq \sum_{j=1}^{\phi_1+\phi_2} \beta'_j t_j.$$

Proof. $\sum_{j=1}^{\phi_1+\phi_2} \beta_j t_j - \sum_{j=1}^{\phi_1+\phi_2} \beta'_j t_j = (\sum_{k=1}^{\phi_1+\phi_2} \beta_k - \sum_{k=1}^{\phi_1+\phi_2} \beta'_k)(t_{\phi_1+\phi_2}) + \sum_{j=1}^{\phi_1+\phi_2-1} (\sum_{k=1}^j \beta_k - \sum_{k=1}^j \beta'_k)(t_j - t_{j+1}) \geq 0$, since each term in parentheses is nonnegative. ■

Lemma 3. Let $\Pi = (\mathcal{T}, \gamma)$ be a feasible solution such that $(e, f) \in A(T)$ and $U_e > U_f$. Then, Π is not optimal.

Proof. Assume for purposes of contradiction that the lemma is untrue, so that an optimal solution $\Pi = (\mathcal{T}, \gamma)$ exists such that $(e, f) \in A(T)$ and $U_e > U_f$. Then, there exists a subtree of $\mathcal{T}$ rooted at some vertex h with the

following structure (see Fig. 3), where h and j_i play the roles of e and f , respectively, in the statement of the lemma:

- (i) h transmits to vertices $j_1, \dots, j_s$ in order, where $s \geq 1$ and $U_h > U_{j_i}$ for some $i \in \{1, \dots, s\}$.
- (ii) Some or all of the j_r 's may be the roots of nontrivial (meaning cardinality ≥ 2) trees $\mathcal{T}(j_r)$ of the broadcast tree.
- (iii) j_i may have immediate successors $k_1, \dots, k_q$ to which it broadcasts in order, and each of these vertices may also be the root of a nontrivial subtree.

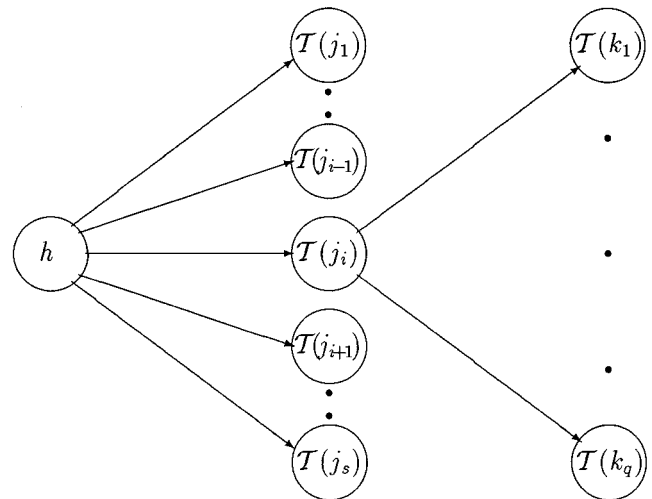


Fig. 3. Initial subtree for Lemma 3.

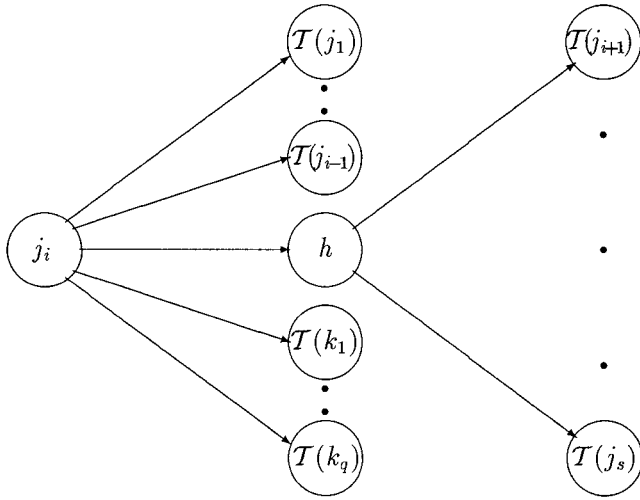


Fig. 4. Altered subtree for Lemma 3.

$\mathcal{T}(h)$ is a subtree of the broadcast tree. We shall replace $\mathcal{T}(h)$ in the broadcast tree by an altered subtree having the same vertex set and show that the altered broadcast tree has a decreased objective function value. Figure 4 illustrates the new subtree:

- (i) j_i transmits in order to $j_1, \dots, j_{i-1}, h, k_1, \dots, k_q$.
- (ii) h transmits in order to $j_{i+1}, \dots, j_s$.
- (iii) The subtrees $\mathcal{T}(j_1), \dots, \mathcal{T}(j_{i-1}), \mathcal{T}(j_{i+1}), \dots, \mathcal{T}(j_s), \mathcal{T}(k_1), \dots, \mathcal{T}(k_q)$ remain intact.

Let b_r represent the reception time for vertex r in the original broadcast tree in Figure 3 and let b'_r denote its reception time in the altered broadcast tree in Figure 4. We obtain the results in Table II.

From Table II and since $U_h > U_{j_i}$, we have $b'_h + b'_{j_i} < b_h + b_{j_i}$ and $b'_g < b_g$ for $g \in \{j_1, \dots, j_{i-1}, j_{i+1}, \dots, j_s, k_1, \dots, k_q\} \equiv \Phi$. Since $b'_g < b_g$ for $g \in \Phi$, it follows that for every $r \in T(b_g), g \in \Phi$ we have $b'_r < b_r$. Hence, we conclude that $\sum_{1 \leq j \leq n} b'_j < \sum_{1 \leq j \leq n} b_j$. ■

Let Π^* be a solution produced by Algorithm ISP1 and let Π be any other feasible solution. Π^* and Π are said to *agree over the set* $\{0, \dots, k\}$ if

- (a) $b_i^* = b_i$ for $1 \leq i \leq k$;
- (b) For $1 \leq i \leq k$, the direct predecessor of i in $\mathcal{T}^*$ is identical to the direct predecessor of i in $\mathcal{T}$.

Lemma 4. Let $\Pi^* = (\mathcal{T}^*, \gamma^*)$ be the feasible solution produced by ISP1, and let $\Pi = (\mathcal{T}, \gamma)$ be any optimal solution. Suppose that Π agrees with Π^* over the set $\{0, \dots, k-1\}$ but not over the set $\{0, \dots, k\}$. Then, there

exists another optimal solution $\Pi' = (\mathcal{T}', \gamma')$ such that Π' agrees with Π^* over the set $\{0, \dots, k\}$.

Proof. Suppose that g transmits to k in the solution Π^* . Since Π^* was produced by Algorithm ISP1, it follows that $\gamma^*(g)$ is of the form $\gamma^*(g) = (h_1, \dots, h_q = k, \dots, h_{\sigma^*(g)})$ for some $q \geq 1$, where $h_i \leq h_{i+1}$ for $1 \leq i \leq \sigma^*(g) - 1$. The agreement assumption implies that $\gamma(g)$ is of the form $\gamma(g) = (h_1, \dots, h_{q-1}, h'_q, \dots, h'_{\sigma(g)})$, that is, $\gamma(g)$ and $\gamma^*(g)$ are identical over their first $q-1$ elements. Two cases are considered:

CASE 1. $\gamma(g) = (h_1, \dots, h_{q-1})$.

In this case, Π' is constructed from Π by moving the subtree $\mathcal{T}(k)$ so that g transmits to $\mathcal{T}(k)$ as g 's last transmission. By Step 2 of ISP1, we have $b'_k = b_k^* \leq b_k$; hence, $b_k - b'_k \geq 0$. Therefore, $b'_j = b_j - (b_k - b'_k) \leq b_j$ for $j \in T(k)$. Also, for $j \notin T(k)$, the reception time has not increased, thus $b'_j \leq b_j$ for $j \notin T(k)$. Summing, we get $\sum_{j=1}^n b'_j \leq \sum_{j=1}^n b_j$.

Since Π was assumed to be optimal, we must have $\sum_{j=1}^n b'_j$ exactly equal to $\sum_{j=1}^n b_j$, and by construction, Π' and Π^* agree over the set $\{0, \dots, k\}$ as claimed.

CASE 2. $\gamma(g) = (h_1, \dots, h_{q-1}, h'_q, \dots, h'_{\sigma(g)})$, where $\sigma(g) > q-1$. By assumption, $h'_q > k$.

In this case, set $m = h'_q$. Two subcases are considered:

SUBCASE 1. $U_k = U_m$.

In this subcase, changing the positions of k and m yields $\Pi' = (\mathcal{T}', \gamma')$ and there is no change in the value of the objective function.

SUBCASE 2. $U_k < U_m$.

Subcase 2 requires a more complex construction. First, we compute the cost associated with any subtree $\mathcal{T}(i)$ of a feasible solution $\Pi = (\mathcal{T}, \gamma)$, where $|T(i)|$ is the cardinality of $T(i)$:

$$\begin{aligned} \sum_{h \in T(i)} b_h &= |T(i)|b_i + \sum_{h \in T(i)} (b_h - b_i) \\ &\equiv |T(i)|b_i + C[\mathcal{T}(i)], \end{aligned} \quad (1)$$

where $C[\mathcal{T}(i)]$ is the ‘‘internal cost’’ of the tree $\mathcal{T}(i)$ ordered according to γ and does not depend on the

TABLE II. Completion times for Lemma 3

Vertex r	b_r	b'_r
h	b_h	$b_h + iU_{j_i}$
j_i	$b_h + iU_h$	b_h
$j_r, 1 \leq r \leq i-1$	$b_h + rU_h$	$b_h + rU_{j_i}$
$j_r, i+1 \leq r \leq s$	$b_h + rU_h$	$b_h + iU_{j_i} + (r-i)U_h$
$k_r, 1 \leq r \leq q$	$b_h + iU_h + rU_{j_i}$	$b_h + (i+r)U_{j_i}$

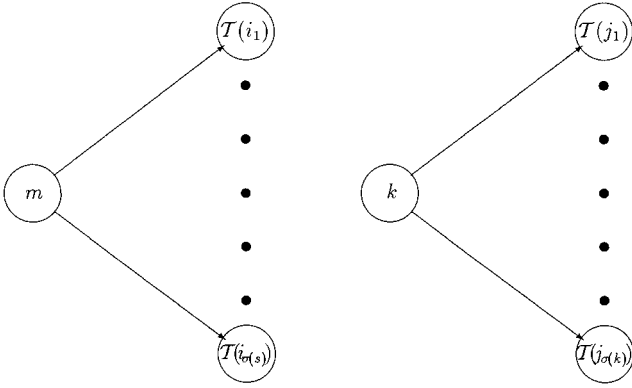


Fig. 5. Initial subtree for Lemma 4.

reception time b_i of vertex i .

Let $\sigma(m) \geq 0$ and $\sigma(k) \geq 0$ denote the number of direct successors of m and k , respectively, in $\mathcal{T}$. We may assume that, in $\mathcal{T}$, vertex m transmits to subtrees $\mathcal{T}(i_1), \dots, \mathcal{T}(i_{\sigma(m)})$ and that k transmits to subtrees $\mathcal{T}(j_1), \dots, \mathcal{T}(j_{\sigma(k)})$, as in Figure 5. Of course, if $\sigma(m) = 0$ or $\sigma(k) = 0$, then the corresponding subtree will not exist.

The reader may find it useful to relate the discussion below to the nonoptimal feasible solution of Figure 6 for Example 2. The notation used in Figure 6 is identical to that in Figure 2. We have $g = 0$, $m = 3$, $k = 1$, $\sigma(m) = 2$, $\sigma(k) = 2$, $(i_1, i_2) = (4, 7)$, and $(j_1, j_2) = (2, 8)$. The construction to be described will convert the feasible solution shown in Figure 6 to that in Figure 2.

We claim that m is not a predecessor of k in $\mathcal{T}$. Otherwise, there would be a path in the broadcast tree $\mathcal{T}$ from m to k , and since $U_m > U_k$, there would necessarily be two adjacent vertices h_1 and h_2 on this path with h_1 transmitting to h_2 , where $U_{h_1} > U_{h_2}$. By Lemma 3, this would mean that Π is not optimal, which contradicts the definition of Π . This proves the claim. Thus, $T(m) \cap T(k) = \emptyset$. Our construction proceeds as follows:

Procedure CONSTRUCT

1. k and m exchange positions: k is located in $\mathcal{T}'$ where m was located in $\mathcal{T}$, and m is located in $\mathcal{T}'$ where k was located in $\mathcal{T}$. Hence, $b'_k = b_m$ and $b'_m = b_k$, so $b'_k + b'_m = b_k + b_m$. In Figure 6, $b'_k = b'_1 = 7$ and $b'_m = b'_3 = 14$.
2. The trees $\mathcal{T}(i_1), \dots, \mathcal{T}(i_{\sigma(m)})$, $\mathcal{T}(j_1), \dots, \mathcal{T}(j_{\sigma(m)})$ remain intact in $\mathcal{T}'$. The root of each of these trees will receive its transmission from either k or m according to Procedure ASSIGN_TREES, described below. The application of this procedure will guarantee that the objective function does not increase.

3. $\mathcal{T}'$ and $\mathcal{T}$ are identical on the vertex set $\Theta_1 = \{0, \dots, n\} \setminus [T(m) \cup T(k)]$. Thus, $\sum_{i \in \Theta_1} b'_i = \sum_{i \in \Theta_1} b_i$. In the example in Figure 6, $\Theta_1 = \{0, 6\}$.

The following procedure assigns the $\sigma(m) + \sigma(k)$ subtrees:

Procedure ASSIGN_TREES

1. The vertex set $\{i_1, \dots, i_{\sigma(m)}, j_1, \dots, j_{\sigma(k)}\}$ is ordered by nonincreasing value of $|T(r)|$ with ties ordered arbitrarily. Let the order so obtained be $(h_1, \dots, h_{\sigma(m) + \sigma(k)})$. In the example in Figure 6, one possible order is $(h_1, \dots, h_4) = (2, 4, 7, 8)$, with corresponding values of $|T(r)|$ being $(2, 1, 1, 1)$.
2. For $1 \leq r \leq \sigma(m) + \sigma(k)$, the tree $\mathcal{T}(h_r)$ is assigned a position in $\mathcal{T}'$ by assigning its root vertex h_r to be the next vertex to receive its transmission either from k or from m , the choice between k and m being made to give the smaller reception time b_{h_r} . Thus, b_{h_r} will equal $b'_k + eU_k$ or $b'_m + eU_m$ for some $e \geq 1$. Terminate.

In our example, after Step 2 of Procedure CONSTRUCT, we have the subtree of Figure 2 which includes only the vertices 0, 1, 3, and 6. After Step 1 of Procedure ASSIGN_TREES is completed, there will follow $4 = \sigma(m) + \sigma(k)$ iterations of Step 2 of ASSIGN_TREES. We proceed as follows: First, we must decide where to assign $\mathcal{T}(h_1) = \mathcal{T}(2)$. Vertex 2 may receive the transmission from either vertex $k = 1$ (reception time would be $12 = 7 + 5$) or from vertex $m = 3$ (reception time would be $21 = 14 + 7$). Hence, we choose vertex 1 to transmit to vertex 2. Continuing similarly, we obtain the solution shown in Figure 2.

The discussion in Steps 1 and 3 of Procedure CONSTRUCT makes it clear that $\sum_{r \in \Theta_1 \cup \{k, m\}} b'_r = \sum_{r \in \Theta_1 \cup \{k, m\}} b_r$. Let $\Theta_2 = \{1, \dots, n\} \setminus [\Theta_1 \cup \{k, m\}]$ = the union of the vertices in the trees $\mathcal{T}(h_r)$, $1 \leq r$

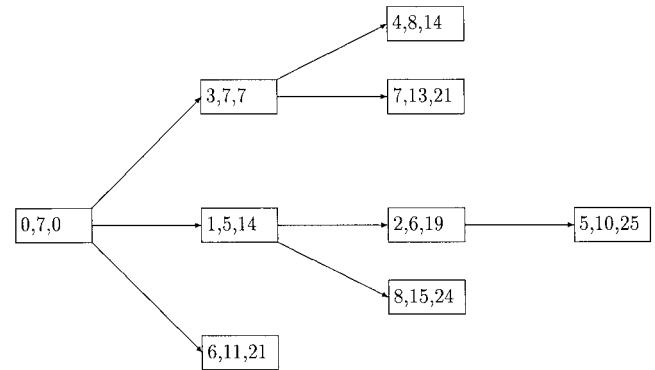


Fig. 6. A nonoptimal broadcast tree for Example 2.

$\leq \sigma(m) + \sigma(k)$. We need to prove that $\sum_{r \in \Theta_2} b_r \geq \sum_{r \in \Theta_2} b'_r$. By relation (1), we need only prove that

$$\sum_{j=1}^{\sigma(m)+\sigma(k)} |T(h_j)| b_{h_j} \geq \sum_{j=1}^{\sigma(m)+\sigma(k)} |T(h_j)| b'_{h_j}. \quad (2)$$

Set the parameters in Lemma 1 as follows: $\delta = b_m$ (which also equals b'_k), $\Delta = U_k$, $\lambda = b_k$ (which also equals b'_m), $\Lambda = U_m$, $\phi_1 = \sigma(m)$, and $\phi_2 = \sigma(k)$.

It follows by construction that $(b'_{h_1}, \dots, b'_{h_{\sigma(m)+\sigma(k)}})$ produced by Procedure ASSIGN_TREES equals $(\beta'_1, \dots, \beta'_{\phi_1+\phi_2})$ of Lemma 1 and that the vector $(b_{h_1}, \dots, b_{h_{\sigma(m)+\sigma(k)}})$ in relation (2) is one particular realization of the vector $(\beta_1, \dots, \beta_{\phi_1+\phi_2})$ in Lemma 1. Hence, the hypotheses of Lemma 1 hold.

To prove (2), we apply Lemma 2 using $t_j = |T(h_j)|$. With these values of t_j , the hypotheses of Lemma 2 hold. The conclusion of Lemma 2 is precisely relation (2). From Step 1 of Procedure CONSTRUCT, we have created an optimal solution Π' that is identical to Π^* over the set $\{0, \dots, k\}$. This completes the proof of Subcase 2 and of Lemma 4. ■

Theorem 2. *Algorithm ISP1 finds an optimal solution to $B|t_{ij} = U_i| \sum_{j=1}^n b_j$ in $O(n \log n)$ time.*

Proof. Optimality follows directly from iterative application of Lemma 4. Algorithm ISP1 enters Step 2 no more than n times. The use of a heap, as described by Kruse [8, p. 364], to maintain the data required for Step 2 enables the required operations to be performed in $O(\log n)$ time. The overall complexity of ISP1 is therefore $O(n \log n)$ time. ■

In general, there may be many optimal solutions, for the following reasons:

- (i) If the U_i 's ($i \geq 1$) are not all distinct, then there is more than one ordering which is monotonically nondecreasing and any such ordering may be used in ISP1.
- (ii) In Step 2 of ISP1, we could choose any p in the set rather than the smallest one.
- (iii) The leaf vertices in any optimal solution may be permuted arbitrarily with no change in the value of the objective function.

3.2. $B|t_{ij} = U_j| \sum_{1 \leq j \leq n} b_j$

An optimal polynomial time algorithm also exists for problem $B|t_{ij} = U_j| \sum_{1 \leq j \leq n} b_j$. As was the case with ISP1, Algorithm ISP2 which accomplishes this is a greedy algorithm, which adds vertices to the broadcast tree in the order $1, \dots, n$ and assigns each vertex k to receive

its transmission from that vertex already in the broadcast tree which will give k the smallest possible reception time. The variables A_i and b_i are interpreted as in the previous section. Note that Algorithms ISP1 and ISP2 differ only in the mechanics of choosing the vertex which yields the smallest reception time.

Algorithm Iterated Shortest Path 2 (ISP2)

Given: positive numbers U_i , $0 \leq i \leq n$.

1. Initialize: $A_0 = \dots = A_n = 0$, $A(T) = \emptyset$, $\gamma_1 = \dots = \gamma_n = \emptyset$.
Order the numbers such that $U_1 \leq \dots \leq U_n$.
2. For $1 \leq i \leq n$
 $L = \min \{A_j | 0 \leq j \leq i-1\}$.
 $p = \min \{j | 0 \leq j \leq i-1 \text{ and } A_j = L\}$.
 $b_i = L + U_i$, $A_i = b_i$, $A_p = b_i$, $A(T) = A(T) \cup \{p, i\}$, $\gamma_p = (\gamma_p, i)$.
 EndFor.

Terminate.

To prove the optimality of ISP2, we need to develop a precise formula for the contribution of U_i to $\sum_{1 \leq k \leq n} b_k$, given a particular feasible solution $\Pi = (\mathcal{T}, \gamma)$. Example 1 (see Fig. 1) helps to illuminate the problem. In this example, each term in the vector $b = (b_1, \dots, b_7)$ consists of a sum of U_j 's, and $\sum_{1 \leq j \leq 7} b_j = 7U_1 + 5U_2 + U_3 + U_4 + 3U_5 + U_6 + U_7$. More generally, for an arbitrary solution $\Pi = (\mathcal{T}, \gamma)$, each b_j is the sum of a number of U_j 's. The question that we consider is how the coefficient of each U_j is related to the structure of the broadcast tree. In Example 1, the coefficient of U_2 in $\sum_{1 \leq j \leq n} b_j$ is 5. From where does this value of 5 come? The transmission from vertex 1 to vertex 2 completes at time $b_2 = b_1 + U_2$. Any successor k of vertex 2 (in Example 1, there is only one such successor, namely, vertex 4) will obviously have U_2 as an addend in the calculation of b_k ; in our example, $b_4 = U_1 + U_2 + U_4$. Moreover, at the same time that vertex 2 becomes available for its transmission to vertex 4, vertex 1 also becomes available for its future transmissions to vertices 5 and 6. Therefore, each vertex $k \in T(5) \cup T(6)$ (this includes vertices 5, 6, and 7) will also have U_2 as an addend in the calculation of U_k . Thus, in the expression for $\sum_{1 \leq j \leq 7} b_j$, the coefficient of 5 for U_2 results from the fact that U_2 contributes to b_2, b_4, b_5, b_6 , and b_7 .

To generalize this discussion based upon Example 1, consider a vertex a which transmits in sequence to trees $\mathcal{T}(j_k)$, $1 \leq k \leq \sigma(a)$ and let $j_h = q$ for some $h \in \{1, \dots, \sigma(a)\}$. Assume that a begins its transmission to q at time t_0 . At the end of the transmission from a to q , both a and q become available for further transmission at time $b_q = t_0 + U_q$. Therefore, U_q will appear as an addend in b_j for all $j \in T(q)$, but also in b_j for j

$\in \cup_{h+1 \leq r \leq \sigma(a)} T(j_r)$, since a 's transmissions to $j_{h+1}, \dots, j_{\sigma(a)}$ follow a 's transmission to q . Letting $Z = \cup_{h \leq r \leq \sigma(a)} T(j_r)$, it follows that the coefficient of U_q in $\sum_{1 \leq j \leq n} b_j$ is $|Z|$. Further,

$$\begin{aligned} \sum_{j \in Z} b_j &= \sum_{j \in Z} [b_q + (b_j - b_q)] \\ &= |Z|b_q + \sum_{j \in Z \setminus \{q\}} [b_j - b_q] \quad (3) \\ &= |Z|[t_0 + U_q] + C(Z \setminus \{q\}), \end{aligned}$$

where $C(Z \setminus \{q\})$ is the ‘‘internal cost’’ related to Z . Each term $b_j - b_q$ in $C(Z \setminus \{q\})$ is a sum of U_k 's for some subset of vertices in $Z \setminus \{q\}$. In Example 1, and setting $q = 2$, we get $Z = \{2, 4, 5, 6, 7\}$, $t_0 = b_1 = U_1$, $U_q = U_2$, and the terms $b_j - b_q = b_j - b_2$ are calculated as follows: $b_4 - b_2 = [U_1 + U_2 + U_4] - [U_1 + U_2] = U_4$, $b_5 - b_2 = [U_1 + U_2 + U_5] - [U_1 + U_2] = U_5$, $b_6 - b_2 = [U_1 + U_2 + U_5 + U_6] - [U_1 + U_2] = U_5 + U_6$, $b_7 - b_2 = [U_1 + U_2 + U_5 + U_7] - [U_1 + U_2] = U_5 + U_7$. Thus, $\sum_{j \in Z} b_j = |Z|(t_0 + U_q) + C(Z \setminus \{q\})$, where $|Z|(t_0 + U_q) = 5(U_1 + U_2)$ and $C(Z \setminus \{q\}) = [U_4] + [U_5] + [U_5 + U_6] + [U_5 + U_7] = U_4 + 3U_5 + U_6 + U_7$. Notice that, as claimed, each term $[b_j - b_q]$ in $C(Z \setminus \{q\})$ is a sum of U_k 's for a subset of vertices in $Z \setminus \{q\}$.

Lemma 5. Let $\Pi^* = (\mathcal{I}^*, \gamma^*)$ be the feasible solution produced by ISP2, and let $\Pi = (\mathcal{I}, \gamma)$ be any other feasible solution. Suppose that Π and Π^* agree over the set of vertices $\{0, \dots, i-1\}$, but not over the set $\{0, \dots, i\}$. Then, there exists a feasible solution $\Pi' = (\mathcal{I}', \gamma')$ which agrees with Π^* over the set $\{0, \dots, i\}$ and such that $\sum_{1 \leq j \leq n} b'_j \leq \sum_{1 \leq j \leq n} b_j$.

Proof. Suppose that g transmits to i in the solution Π^* . Since Π^* was produced by Algorithm ISP2, it follows that $\gamma^*(g)$ is of the form $\gamma^*(g) = (h_1, \dots, h_q = k, \dots, h_{\sigma^*(g)})$ for some $q \geq 1$, where $h_j \leq h_{j+1}$ for $1 \leq j \leq \sigma^*(g) - 1$. Also, Algorithm ISP2 yields $b_i \geq b_i^*$. The agreement assumption implies that $\gamma(g)$ is of the form $\gamma(g) = (h_1, \dots, h_{q-1}, h'_q, \dots, h'_{\sigma(g)})$, that is, $\gamma(g)$ and $\gamma^*(g)$ are identical over their first $q-1$ elements.

CASE 1. $\gamma(g) = (h_1, \dots, h_{q-1})$.

Each j (if any) that is a direct successor of g in Π has the property that $j \leq i-1$.

We create Π' by a two-step conversion from Π . First, $\mathcal{I}(i)$ is removed from Π to form $\Pi'' = (\mathcal{I}'', \gamma'')$, the restriction of Π to $N \setminus T(i)$. Clearly, for each $j \in T'' = N \setminus T(i)$, $b''_j \leq b_j$. Second, $\mathcal{I}(i)$ is attached to the solution Π'' by having vertex $g \in T''$ transmit to vertex i , with i being the last vertex to which g transmits. Thus, $(g, i) \in A(T')$ and $\mathcal{I}'(i) = \mathcal{I}(i)$. Since $b'_i = b_i^*$, it follows that $b'_i \leq b_i$ and, hence,

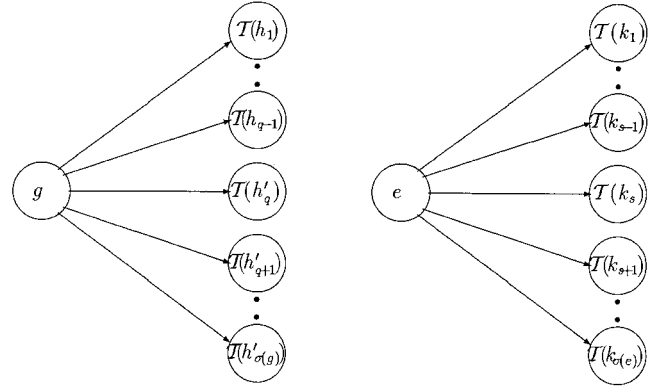


Fig. 7. Lemma 6, Case 2, with $\mathcal{I}(h'_q) = \mathcal{I}(m)$ and $\mathcal{I}(k_s) = \mathcal{I}(i)$.

that $b'_j \leq b_j$ for $j \in T(i) = T'(i)$. Thus, $\sum_{1 \leq j \leq n} b'_j \leq \sum_{1 \leq j \leq n} b_j$ as claimed.

CASE 2. $\gamma(g) = (h_1, \dots, h_{q-1}, h'_q, \dots, h'_{\sigma(g)})$, where $\sigma(g) > q-1$. By assumption, $h'_q > i$.

To simplify the exposition, we let $m = h'_q$. There exists some vertex e which transmits to i in Π . We may assume that $\gamma(e) = (k_1, \dots, k_s = i, \dots, k_{\sigma(e)})$. The relationships of g and e to their successor vertices are illustrated in Figure 7. The possibility that $e = g$ is not excluded; if $e = g$, then $k = h'_s$ for some $s > q$.

Let $Z_m = \cup_{q \leq j \leq \sigma(g)} T(h'_j)$ and $Z_i = \cup_{s \leq j \leq \sigma(e)} T(k_j)$. As discussed prior to the statement of the lemma, $|Z_m|$ and $|Z_i|$ are the coefficients (multipliers) of U_m and U_i , respectively, in $\sum_{1 \leq j \leq n} b_j$.

SUBCASE 1. $|Z_m| \geq |Z_i|$.

In this subcase, we shall retain the solution $\Pi = (\mathcal{I}, \gamma)$ but alter the instance $(U_0, \dots, U_n)$ in a two-step procedure. Let $\Delta = U_m - U_i$. First, substitute $U'_i = U_i + \Delta$ for U_i . From relation (3), this increases the objective function by $\Delta|Z_i|$. Next, substitute $U'_m = U_m - \Delta$ for U_m . Again using (3), this decreases the objective function by $\Delta|Z_m|$. The two-step conversion above yields a net change of $\Delta(|Z_i| - |Z_m|) \leq 0$ in the objective function. This conversion is equivalent to exchanging the positions of i and m in $\Pi = (\mathcal{I}, \gamma)$ to yield a new solution $\Pi' = (\mathcal{I}', \gamma')$ which agrees with Π^* on the set $\{0, \dots, i\}$ and such that $\sum_{1 \leq j \leq n} b'_j \leq \sum_{1 \leq j \leq n} b_j$. Thus, Π' is the solution specified by the lemma.

SUBCASE 2. $|Z_m| < |Z_i|$. Then, $i \notin Z_m$ (since $i \in Z_m$ implies $|Z_i| < |Z_m|$).

In this subcase, we form $\Pi' = (\mathcal{I}', \gamma')$ by interchanging the positions of the vertices in Z_m and in Z_i , as illustrated in Figure 8, so that g transmits in order to $\mathcal{I}(h_1), \dots, \mathcal{I}(h_{q-1}), \mathcal{I}(k_s) = \mathcal{I}(i), \mathcal{I}(k_{s+1}), \dots, \mathcal{I}(k_{\sigma(e)})$ and e transmits in order to $\mathcal{I}(k_1), \dots,$

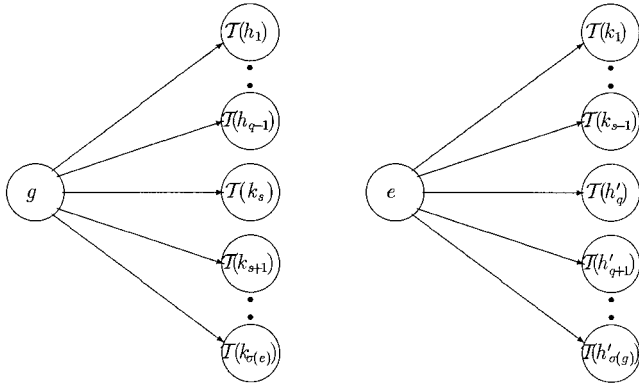


Fig. 8. Lemma 6, Case 2, Subcase 2, with $\mathcal{T}(k_s) = \mathcal{T}(i)$ and $\mathcal{T}(h'_q) = \mathcal{T}(m)$.

$\mathcal{T}(k_{s-1}), \mathcal{T}(h'_q) = \mathcal{T}(m), \mathcal{T}(h'_{q+1}), \dots, \mathcal{T}(h'_{\sigma(g)})$. The cost associated with the vertices in $Z_m \cup Z_i$ before the interchange takes place can be calculated from (3), where t_0 and t'_0 are the times at which m and i , respectively, begin reception of the message in solution Π :

$$A = |Z_m|(t_0 + U_m) + C(Z_m \setminus \{m\}) \\ + |Z_i|(t'_0 + U_i) + C(Z_i \setminus \{i\}).$$

The cost associated with the vertices in $Z_1 \cup Z_2$ in Π' (i.e., after the interchange) is

$$B = |Z_i|(t_0 + U_i) + C(Z_i \setminus \{i\}) \\ + |Z_m|(t'_0 + U_m) + C(Z_m \setminus \{m\}).$$

Thus, $A - B = (|Z_i| - |Z_m|)(t'_0 - t_0)$. In the subcase under consideration, $|Z_i| - |Z_m| > 0$. Now, if $t'_0 < t_0$, then e begins transmission to i before g begins transmission to m in the solution Π . This implies that g was chosen incorrectly to transmit to i in Step 2 of the solution Π^* produced by ISP2, a contradiction. Thus, $t'_0 \geq t_0$ which implies that $A - B \geq 0$ and, hence, that Π' satisfies the conditions of the lemma. This completes the proof of the lemma. ■

Theorem 3. Algorithm ISP2 finds an optimal solution to $B|t_{ij} = U_j|\sum_{1 \leq j \leq n} b_j$ in $O(n \log n)$ time.

Proof. That ISP2 finds an optimal solution follows from Corollary 1 and iterative application of Lemma 5. The U_j 's may be ordered in $O(n \log n)$ time. By using a heap to store the data for Step 2, as in ISP1, Algorithm ISP2 can be implemented in $O(n \log n)$ time. ■

Remark 1. One of the many optimal schedules available from ISP2 requires processor i , $0 \leq i \leq n$, to transmit

in ascending index order to all processors $(2i + 1)2^k$ for those values of $k \geq 0$ such that $(2i + 1)2^k \leq n$.

Corollary 2. Algorithm ISP2 finds an optimal solution to $B|t_{ij} = \max\{U_i, U_j\}|\sum b_j$ in $O(n \log n)$ time.

Proof. The solution produced by Algorithm ISP2 has the property that all transmissions are made from a lower numbered vertex to a higher numbered vertex, that is, each arc $(i, j) \in A(T)$ has the property that $i < j$. This implies that $\max\{U_i, U_j\} = U_j$ for each transmission. Hence, the solution produced by ISP2 will have the same objective function value for $B|t_{ij} = \max\{U_i, U_j\}|\sum_{1 \leq j \leq n} b_j$ as for $B|t_{ij} = U_j|\sum_{1 \leq j \leq n} b_j$. Since the optimal value for the $t_{ij} = U_j$ problem is a lower bound on that for the $t_{ij} = \max\{U_i, U_j\}$ problem, it follows that ISP2 finds an optimal solution for the problem with $t_{ij} = \max\{U_i, U_j\}$. ■

3.3. $B|t_{ij} = U_j|b^t$

We describe a polynomial time algorithm for the problem of minimizing the broadcast cycle time when transmission times depend only upon the receiving vertex. We begin with some preliminaries. Let v be a vertex of a broadcast tree $\mathcal{T} = (T, A(T))$, and recall that $s(v) =$ the successor set of $v = \{u | (v, u) \in A(T)\}$. Vertex v is busy for U_v time units during which it receives the message and $\sum_{m \in s(v)} U_m$ time units during which it transmits the message. Thus, $c_v =$ the busy time of $v = U_v + \sum_{m \in s(v)} U_m$. Finally, $b^t = \max\{c_v | 0 \leq v \leq n\}$.

Define a chain to be a rooted tree in which each vertex has precisely one successor, except for a single leaf vertex. Thus, for example, $\mathcal{T} = (T, A(T))$, where $T = \{0, 1, 2, 3, 4, 5\}$ and $A = \{(0, 5), (5, 1), (1, 4), (4, 2), (2, 3)\}$, may also be denoted by the chain notation $\{0, 5, 1, 4, 2, 3\}$. Recall that in this paper “transitive arcs” such as $(0, 1), (0, 2), \dots$ are not included in A . Corresponding to this chain would be the solution $\Pi = (\mathcal{T}, \gamma)$, where $\gamma = (\gamma_0, \gamma_1, \gamma_2, \gamma_3, \gamma_4, \gamma_5) = ((5), (4), (3), \emptyset, (2), (1))$. The algorithm below produces optimal solutions which are chains; there may be other optimal solutions as well.

Algorithm High–Low Chain (HLC)

Given: positive numbers U_i , $0 \leq i \leq n$.

1. Initialize: $A(T) = \emptyset$, $\gamma_1 = \dots = \gamma_n = \emptyset$.
Order the vertices such that $U_1 \leq \dots \leq U_n$.
2. Form the $n + 1$ vertices into a chain in the sequence $\{0, n, 1, n - 1, 2, n - 2, \dots, n/2 + 1, n/2\}$ if n is even, or $\{0, n, 1, n - 1, 2, n - 2, \dots, \lfloor n/2 \rfloor, \lfloor n/2 \rfloor + 1\}$ if

n is odd.

Update $A(T)$ and γ accordingly.

Terminate.

To prove the optimality of Algorithm HLC, we first prove two results about directed rooted trees. Two vertices, i and j , of a broadcasting tree are said to be *near* each other if one is a direct predecessor of the other or if there exists a third vertex, k , which is a direct predecessor of both of them.

Lemma 6. *If i and j are near each other, then $b^t \geq U_i + U_j$.*

Proof. Suppose that processors i and j in a broadcast tree are near each other. If one (say, i) is a direct predecessor of the other (say, j), then $c_i = U_i + \sum_{m \in s(i)} U_m \geq U_i + U_j$. On the other hand, if there exists k such that $i, j \in s(k)$, then $c_k = U_k + \sum_{m \in s(i)} U_m \geq U_i + U_j$. ■

Lemma 7. *Let $\mathcal{T} = (T, A(T))$ be a rooted tree that is oriented away from the root and consists of $n + 1$ vertices labeled from 0 to n , with the root vertex having the label 0. Then, for each i , $1 \leq i \leq \lfloor n/2 \rfloor$, there exist indices i' and j' such that*

- (1) $i' \geq i$ and $j' \geq n + 1 - i$, and
- (2) i' and j' are near each other.

Proof. Assume that there exists an index i^* , where $1 \leq i^* \leq \lfloor n/2 \rfloor$, for which there do not exist indices i' and j' such that $i' \geq i^*$, $j' \geq n + 1 - i^*$, and i' and j' are near each other. Based upon this assumption, we shall derive a contradiction:

Define $X = \{i | i^* \leq i \leq n - i^*\}$, $Y = \{j | n - i^* + 1 \leq j \leq n\}$, $X' = \{i_0 | i_0 \text{ is a direct predecessor of at least one } i \in X\}$, and $Y' = \{j_0 | j_0 \text{ is a direct predecessor of at least one } j \in Y\}$. From the definition of i^* , we get observations (a) and (b) below:

- (a) For any $i \in X \cup Y$ and $j \in Y$, i and j are not near each other. This has several implications: $X \cap Y' = \emptyset$, $X' \cap Y = \emptyset$, $X' \cap Y' = \emptyset$, and $Y \cap Y' = \emptyset$. Also, no two elements of Y have the same direct predecessor, so $|Y| = |Y'| = i^*$.
- (b) Since $0 \notin X$, there must exist $h \notin X$ and $i \in X$ such that h is a direct predecessor of i . Hence, $|X \cup X'| \geq |X| + 1 = n - 2i^* + 2$.
- (c) By definition, $X \cap Y = \emptyset$.

Observations (a)–(c) yield

$$n + 1 = |T| \geq |X \cup X' \cup Y \cup Y'|$$

$$= |X \cup X'| + |Y| + |Y'|$$

$$\geq (n - 2i^* + 2) + i^* + i^* = n + 2,$$

a contradiction. ■

Theorem 4. *Algorithm HLC finds an optimal solution to $B|t_{ij} = U_j|b^t$ in $O(n \log n)$ time.*

Proof. The theorem is proved below for the case of n odd. The proof for n even is almost identical. For n odd, the chain is $\{0, n, 1, n - 1, \dots, \lfloor (n/2) \rfloor, \lfloor (n/2) \rfloor + 1\}$. Each $i \neq 0$ receives the message in U_i time units, and each $i \neq \lfloor (n/2) \rfloor + 1$ transmits the message to the subsequent vertex. The busy time vector $(c_0, \dots, c_n)$ is thus easily calculated: $c_0 = U_n$, $c_i = U_i + U_{n-i}$ for $1 \leq i \leq \lfloor (n/2) \rfloor$, $c_{\lfloor (n/2) \rfloor + 1} = U_{\lfloor (n/2) \rfloor + 1}$, and $c_i = U_i + U_{n+1-i}$ for $\lfloor (n/2) \rfloor + 2 \leq i \leq n$. It is easy to verify that $c_0 \leq c_n$, $c_{\lfloor (n/2) \rfloor + 1} \leq c_{\lfloor (n/2) \rfloor}$ and $c_i \leq c_{n+1-i}$ for $1 \leq i \leq \lfloor (n/2) \rfloor$. Thus, for our chain solution,

$$b^t = \max \{c_0, \dots, c_n\}$$

$$= \max \{U_i + U_{n+1-i} | 1 \leq i \leq \lfloor (n/2) \rfloor\}.$$

Let the maximum in this last expression be assumed for $i = i^*$. Then, $b^t = U_{i^*} + U_{n+1-i^*}$.

To prove that the chain solution is optimal, it suffices to show that $U_{i^*} + U_{n+1-i^*}$ is a lower bound on b^t for any solution $\Pi = (\mathcal{T}, \gamma)$. In any such solution $\Pi = (\mathcal{T}, \gamma)$, $\mathcal{T}$ satisfies the hypotheses of Lemma 7, so there exist indices $i' \geq i^*$ and $j' \geq j^*$ such that i' and j' are near each other in $\mathcal{T}$. Then, by Lemma 6, $U_{i^*} + U_{n+1-i^*}$ is a lower bound on the cycle time for *all* broadcast trees. This completes the proof of optimality. The dominant computational step in HLC is ordering the vertices in Step 1, which requires $O(n \log n)$ time. ■

It is worth noting that Step 2 of Algorithm HLC requires only linear time. Therefore, if $U_1, \dots, U_n$ are given already ordered, then the time complexity is $O(n)$.

Remark 2. *The same cycle time is achieved by reversing the order of vertices $1, \dots, n$ in the chain found by HLC. This provides additional flexibility in the design of the broadcast system. Note that many other optimal solutions may exist as well.*

4. COMPUTATIONAL COMPLEXITY RESULTS

In this section, we provide proofs of unary *NP*-completeness for several of the broadcasting problems described in Section 2. In each case, it is easy to see that the recogni-

tion version of the problem being considered is in *NP*. We use the convention that km vertices with parameter p_j denotes k copies of a vertex with parameter p_j , $j = 1, \dots, m$. Unless otherwise stated, we assume $m = 2^k$ for some integer k , since this can be enforced by adding suitable “dummy” data without doubling the size of the problem. Also, we let $M = \sum_{i=1}^m 2(x_i + y_i + z_i) + 1$.

4.1. Sum of Broadcast Times Problems

Here, we attempt to minimize the total broadcast time to all vertices, given that the transmission time between a pair of vertices is an arbitrary function of the parameters of those vertices.

Theorem 5. *The recognition version of $B|_{t_{ij}} = f(U_i, U_j) | \sum_{j=1}^n b_j$ is unary NP-complete.*

Proof. By reduction from the following problem known to be unary NP-complete. Exact Cover by 3-Sets [3, p. 221]: Given a $3m \times 3m$ zero-one matrix $A = \{a_{ij}\}$ with each column sum equal to three, does there exist a subset of m columns in which each row has exactly one entry of 1?

Here, we do not assume that $m = 2^k$ for some integer k . Denote by e_{ij} the $i \times j$ matrix consisting entirely of ones. Consider the following instance of $B|_{t_{ij}} = f(U_i, U_j) | \sum_{j=1}^n b_j$:

$n = 6m =$ the number of vertices, not including the initiator (vertex 0).

$C = 6m^3 + 3m^2 + 6m$ denotes the threshold cost.

The matrix $f = f(U_i, U_j)$ is given below, for $0 \leq i, j \leq n$, where 0 denotes the initiator:

$$\begin{pmatrix} (C+1)e_{1,1} & me_{1,3m} & (C+1)e_{1,3m} \\ (C+1)e_{3m,1} & (C+1)e_{3m,3m} & A' \\ (C+1)e_{3m,1} & (C+1)e_{3m,3m} & (C+1)e_{3m,3m} \end{pmatrix},$$

where $A' = (a'_{ij})$ is defined by $a'_{ij} = 1$ if $a_{ji} = 1$, $a'_{ij} = C + 1$ otherwise.

We prove that there exists a solution to this instance of $B|_{t_{ij}} = f(U_i, U_j) | \sum_{j=1}^n b_j$ with $\sum_{j=1}^n b_j \leq C$ if and only if there exists a solution to Exact Cover by 3-Sets.

($\Rightarrow$) Figure 9 shows a network in which $\sum_{j=1}^n b_j = C$. Numbers on arcs represent the order of transmission from a vertex, as in Figure 1, where without loss of generality the first m columns of A provide a solution to Exact Cover by 3-Sets.

($\Leftarrow$) Observe from matrix f that, if the cost of C is not to be exceeded, the vertices $3m + 1, \dots, 6m$ can only be reached through the vertices $1, \dots, 3m$. Moreover, since each vertex from $1, \dots, 3m$ can reach only three vertices from $3m + 1, \dots, 6m$, it will be necessary

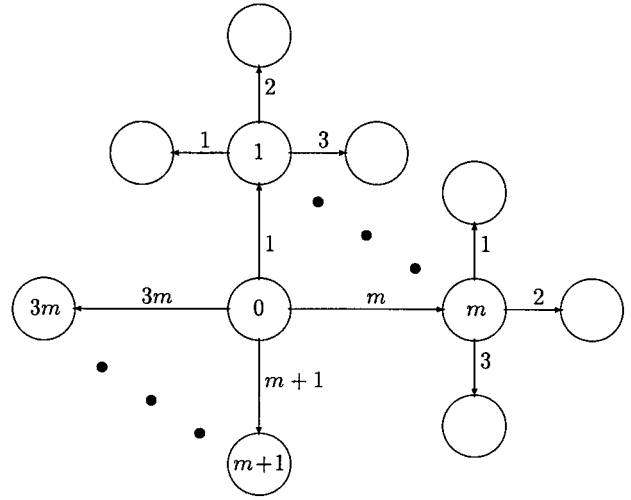


Fig. 9. Proof of Theorem 5.

to use at least m of the vertices from $1, \dots, 3m$ to reach the other vertices. The minimum total transmission time to vertices $1, \dots, 3m$ is $m + 2m + \dots + 3m^2 = (9m^3 + 3m^2)/2$. A lower bound on the total transmission time to the other vertices, since they must be reached through vertices $1, \dots, 3m$, and no more than three can be reached from each, is given by $3(m + \dots + m^2) = (3m^3 + 3m^2)/2$, exclusive of transmission time from the vertex in $1, \dots, 3m$ to be used. Now, $C - (9m^3 + 3m^2)/2 - (3m^3 + 3m^2)/2 = 6m$. This implies that each of the first m vertices reached from vertex 0 must transmit to three vertices from $3m + 1, \dots, 6m$ in a total of $1 + 2 + 3 = 6$ time units. This, in turn, implies the existence of a solution to Exact Cover by 3-Sets. ■

The presence of nonzero ready times for vertices complicates the minimization of total broadcast time in the case where transmission times depend only upon the sending vertex.

Theorem 6. *The recognition version of $B|_{t_{ij}} = U_i, r_j | \sum_{j=1}^n b_j$ is unary NP-complete.*

Proof. See the Appendix for a summary and Hall et al. [4] for the details. ■

Theorem 7. *The recognition version of $B|_{t_{ij}} = U_j, r_j | \sum_{j=1}^n b_j$ is unary NP-complete.*

The proof is summarized in the Appendix. ■

Corollary 3. *The recognition version of $B|_{t_{ij}} = \max\{U_i, U_j\}, r_j | \sum_{j=1}^n b_j$ is unary NP-complete.*

Proof. Follows from Theorem 7, since transmission is always made to a receiving vertex with a U_j value no smaller than that of the sending vertex. ■

Corollary 4. The recognition version of $B|t_{ij} = f(U_i, U_j)$, $r_j|\sum_{j=1}^n b_j$ is unary NP-complete.

Proof. Follows immediately from the fact that $B|t_{ij} = U_j$, $r_j|\sum_{j=1}^n b_j$ is a special case and from Theorem 7. ■

4.2. Weighted Sum of Broadcast Time Problems

We discuss the problem of minimizing the total weighted broadcast time, in the case where transmission time depends only upon the sending vertex.

Theorem 8. The recognition version of $B|t_{ij} = U_i|\sum_{j=1}^n w_j b_j$ is unary NP-complete.

Proof. By reduction from Numerical 3-Dimensional Matching (as defined in the proof of Theorem 6 in the Appendix). Instance of $B|t_{ij} = U_i|\sum_{j=1}^n w_j b_j$:

$$n = 16m - 1.$$

We have the following vertices:

- (R_0) m vertices, including vertex 0, with $U_j = 8M$, $w_j = 8M$
- (R_1) m vertices with $U_j = 9M + x_j$, $w_j = 4M$
- (R_2^a) m vertices with $U_j = 10M + y_{\max}$, $w_j = 2M + y_{\max}$
- (R_2^b) m vertices with $U_j = 10M + y_j$, $w_j = 2M + y_j$
- (R_3^a) $3m$ vertices with $U_j = 11M$, $w_j = M + z_{\max}$
- (R_3^b) m vertices with $U_j = 11M$, $w_j = M + z_j$
- (R_4) $8m$ vertices with $U_j = C + 1$, $w_j = 1$.

$$\begin{aligned}
 C = M^2[& 64 \sum_{j=1}^k j 2^{j-1} + 4m(8k + 8) + 2m(8k + 16) \\
 & + 2m(8k + 17) + m(8k + 24) + m(8k + 26) \\
 & + m(8k + 26) + m(8k + 27)] \\
 & + M[m(8k + 16)y_{\max} + (8k + 18) \sum_{j=1}^m y_j \\
 & + my_{\max} + 5 \sum_{j=1}^m x_j + m(24k + 76)z_{\max} \\
 & + (8k + 27) \sum_{j=1}^m z_j + m(64k + 287)] \\
 & + [7 \sum_{j=1}^m x_j + 3 \sum_{j=1}^m y_j + 3my_{\max} \\
 & + (2 \sum_{j=1}^m x_j + my_{\max})z_{\max}] + mb^2/2 \\
 & - \sum_{j=1}^m (x_j^2 + y_j^2 + z_j^2)/2.
 \end{aligned}$$

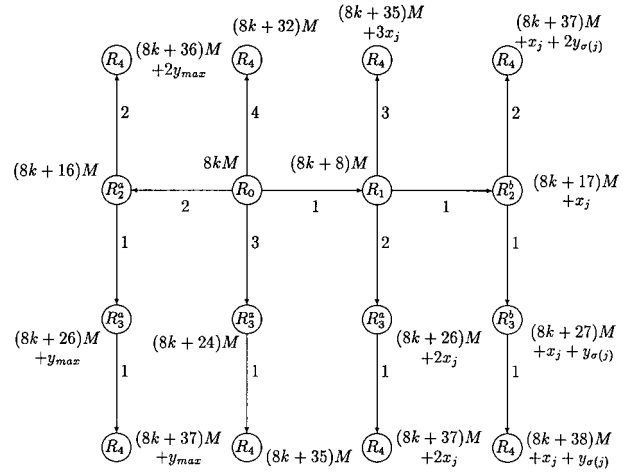


Fig. 10. One of m subtrees in the proof of Theorem 8.

($\Rightarrow$) Figure 10 shows a network in which $\sum_{j=1}^n w_j b_j = C$ (note that $m = 2^k$).

($\Leftarrow$) Because of their high U_j values, the vertices in R_4 will be leaves in any schedule with cost $\leq C$. It follows from the weight-to-processing time ratio rule of Smith [13] that the optimal order of transmission to the various subsets of vertices will be, first, R_0 , then R_1 , then R_2^a and R_2^b , then R_3^a and R_3^b , and, finally, R_4 . The M terms in the broadcast times of the 14 vertices excluding R_0 and R_1 in Figure 10 have coefficients, ignoring the $8k$ term in common, of 16, 17, 24, 26, 26, 27, 32, 35, 35, 36, 37, 37, 37, and 38. Since M is much larger than the other terms, and smaller broadcast times will be associated with larger weights in an optimal solution, the subset to which each of those vertices belongs is (respectively) R_2^a , R_2^b , $R_3^a(3)$, R_3^b , and $R_4(8)$. Thus, any schedule with cost of C must have the same cost components as the schedule shown in Figure 10. Assume that vertex $j \in R_1$ transmits to vertex $\sigma(j) \in R_2^b$, which, in turn, transmits to vertex $\phi(j) \in R_3^b$. The cost of this schedule can be computed as $C - mb^2/2 + \sum_{j=1}^m (x_j^2 + y_j^2 + z_j^2)/2 + \sum_{j=1}^m (x_j y_{\sigma(j)} + x_j z_{\phi(j)} + y_{\sigma(j)} z_{\phi(j)}) = C - mb^2/2 + \sum_{j=1}^m (x_j + y_{\sigma(j)} + z_{\phi(j)})^2/2 \geq C$, where equality is only attained if $x_j + y_{\sigma(j)} + z_{\phi(j)} = b$ for $j = 1, \dots, m$, which implies the existence of a solution to Numerical 3-Dimensional Matching. ■

Next, we consider the case where transmission time depends only upon the receiving vertex.

Theorem 9. The recognition version of $B|t_{ij} = U_j|\sum_{j=1}^n w_j b_j$ is unary NP-complete.

Proof. See the Appendix for a summary and [4] for the details. ■

Corollary 5. The recognition version of $B|t_{ij} = \max\{U_i, U_j\}|\sum_{j=1}^n w_j b_j$ is unary NP-complete.

Proof. This follows from Theorem 9, using the argument of Corollary 3. ■

Corollary 6. *The recognition version of $B|_{t_{ij} = f(U_i, U_j)} | \sum_{j=1}^n w_j b_j$ is unary NP-complete.*

Proof. This follows immediately from the fact that $B|_{t_{ij} = U_j} | \sum_{j=1}^n w_j b_j$ is a special case and from Theorem 9. ■

4.3. Maximum Broadcast Time Problems

Theorem 10. *The recognition version of $B|_{t_{ij} = U_i} | \max_{1 \leq j \leq n} \{b_j\}$ is unary NP-complete.*

Proof. See the Appendix for a summary and [4] for the details. ■

Theorem 11. *The recognition version of $B|_{t_{ij} = U_j} | \max_{1 \leq j \leq n} \{b_j\}$ is unary NP-complete.*

The proof is summarized in the Appendix. ■

Corollary 7. *The recognition version of $B|_{t_{ij} = \max\{U_i, U_j\}} | \max_{1 \leq j \leq n} \{b_j\}$ is unary NP-complete.*

Proof. This follows from Theorem 11, using the argument of Corollary 3. ■

Corollary 8. *The recognition version of $B|_{t_{ij} = f(U_i, U_j)} | \max_{1 \leq j \leq n} \{b_j\}$ is unary NP-complete.*

Proof. This follows immediately from the fact that $B|_{t_{ij} = U_j} | \max_{1 \leq j \leq n} \{b_j\}$ is a special case and from Theorem 11. ■

4.4. Broadcast Cycle Time Problems

Theorem 12. *The recognition version of $B|_{t_{ij} = \max\{U_i, U_j\}} | b^t$ is unary NP-complete.*

Proof. By reduction from the following problem which is known to be unary NP-complete. Numerical Matching with Target Sums [3, p. 224]: Given vectors $X = \{x_1, \dots, x_m\}$, $Y = \{y_1, \dots, y_m\}$, and $Z = \{z_1, \dots, z_m\}$, does there exist a partition of $X \cup Y \cup Z$ into m subsets $S_1, \dots, S_m$ such that S_i contains one element from each of X , Y and Z , $i = 1, \dots, m$ and such that for $i = 1, \dots, m$ if $S_i = \{x_j, y_k, z_l\}$, then $x_j = y_k + z_l$?

Consider an instance of $B|_{t_{ij} = \max\{U_i, U_j\}} | b^t$ defined as follows:

$$n = 6m + 1$$

- (R_0) 1 vertex with $U_0 = 1$
- (R_1) m vertices with $U_j = x_j$
- (R_2) m vertices with $U_j = M + y_j$
- (R_3) m vertices with $U_j = M^2 + z_j$

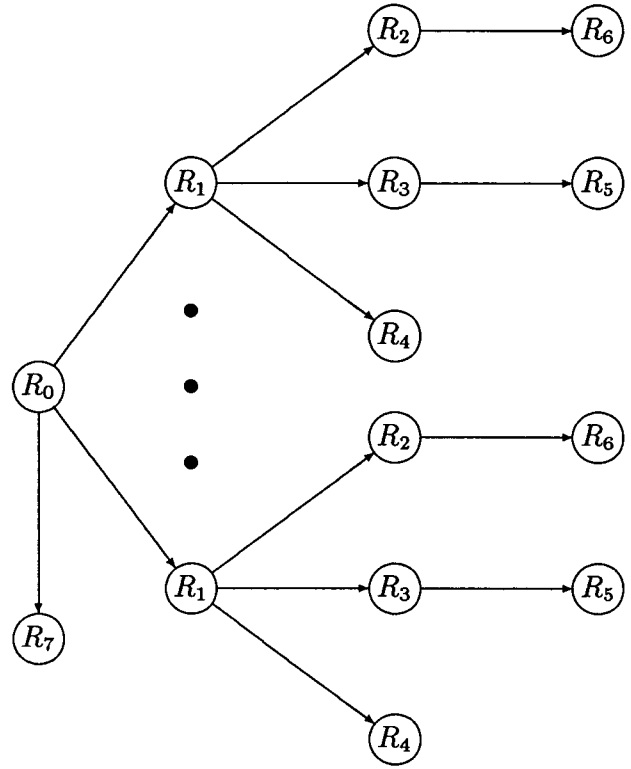


Fig. 11. Proof of Theorem 12.

- (R_4) m vertices with $U_j = C - (M^2 + M + 2x_j)$
- (R_5) m vertices with $U_j = C - (M^2 + z_j)$
- (R_6) m vertices with $U_j = C - (M + y_j)$
- (R_7) 1 vertex with $U_j = C - \sum_{h=1}^m x_h$
- $C = M^3$.

($\Rightarrow$) Figure 11 shows a network in which the cycle time of any vertex $j \in R_2$ is $[C - (M + y_j)] + M + y_j = C$. It follows that $b^t = C$. Because of the cyclic nature of this problem, the order of transmissions from a vertex is irrelevant.

($\Leftarrow$) We first prove six claims about a hypothetical schedule that achieves a cycle time of C :

Claim 1. All vertices in $R_4 \cup R_5 \cup R_6 \cup R_7$ are leaves.

Proof of Claim 1. If not, at least one vertex from $R_4 \cup R_5 \cup R_6 \cup R_7$ transmits to another vertex and the cycle time of the first vertex exceeds C .

Claim 2. Each vertex in $R_0 \cup R_1 \cup R_2 \cup R_3$ transmits to exactly one vertex from outside that group.

Proof of Claim 2. From Claim 1, there are $3m + 1$ vertices that are leaves. Any other vertex can transmit to at most one of them without exceeding the cycle time of C . Since only $3m + 1$ vertices are not already designated as leaves, the result follows.

Claim 3. Vertex R_7 is a direct successor of either vertex R_0 or a vertex in R_1 .

Proof of Claim 3. If not, the direct predecessor of R_7 has a cycle time exceeding C .

Claim 4. A lower bound on the total cycle time of all vertices in $R_0 \cup R_1 \cup R_2 \cup R_3$ is $(3m + 1)C$.

Proof of Claim 4. Each vertex $j \in R_1 \cup R_2 \cup R_3$ requires a time of at least U_j to receive transmission. Each such reception increases the total cycle time by $2U_j$, since both the sending and receiving vertices are in $R_0 \cup R_1 \cup R_2 \cup R_3$. Also, each vertex $j \in R_0 \cup R_1 \cup R_2 \cup R_3$ has a unique successor $k \in R_4 \cup R_5 \cup R_6 \cup R_7$, adding U_k to the total cycle time. Therefore, the total cycle time is at least $2 \sum_{j=1}^m x_j + 2 \sum_{j=1}^m (M + y_j) + 2 \sum_{j=1}^m (M^2 + z_j) + \sum_{j=1}^m (C - M^2 - M - 2x_j) + \sum_{j=1}^m (C - M^2 - z_j) + \sum_{j=1}^m (C - M - y_j) + C - \sum_{j=1}^m x_j = (3m + 1)C$.

Claim 5. The cycle time of each vertex $j \in R_0 \cup R_1 \cup R_2 \cup R_3$ equals C .

Proof of Claim 5. Since $|R_0 \cup R_1 \cup R_2 \cup R_3| = 3m + 1$, this follows from Claim 4.

Claim 6. Each vertex in R_2 has a unique successor in R_6 , and each vertex in R_3 has a unique successor in R_5 .

Proof of Claim 6. This follows from Claim 5.

We now prove the remainder of the theorem. There are two cases (from Claim 3):

CASE 1. R_7 is a direct successor of R_0 . Here, all the vertices in R_1 must also follow R_0 , in order to give R_0 a cycle time of exactly C , as required by Claim 5. Thus, each vertex in R_4 must receive transmission from a vertex in R_1 , and the schedule is identical to that in Figure 11. The cycle time at vertex $j \in R_1$ only equals C if the sum of the U_j values for the three vertices to which j transmits equals $C - x_j$, $j = 1, \dots, m$, which implies the existence of a solution to Numerical Matching with Target Sums.

CASE 2. R_7 is a direct successor of a vertex $j \in R_1$. If vertex j is to have a cycle time of exactly C , as required by Claim 5, all other vertices in R_1 must be direct successors of vertex j . Similarly, vertex R_0 must transmit to one vertex each from R_2 , R_3 , and R_4 . It follows from Claim 5 that each vertex $k \neq j \in R_1$ must transmit to one vertex in each of R_2 , R_3 , and R_4 . Again, a schedule with maximum cycle time of C is only possible if there exists a solution to Numerical Matching with Target Sums. ■

Corollary 9. The recognition version of $B|t_{ij} = f(U_i, U_j)|b^t$ is unary NP-complete.

Proof. This follows immediately from the fact that $B|t_{ij} = \max\{U_i, U_j\}|b^t$ is a special case and from Theorem 12. ■

5. HEURISTICS

Here, we describe heuristics for two of the problems in Table I and analyze their worst-case performance.

5.1. $B|t_{ij} = U_j|\max_{1 \leq j \leq n}\{b_j\}$

The recognition version of this problem was shown to be unary NP-complete in Theorem 11. This motivates the design and analysis of heuristic solution procedures. We propose the use of Algorithm **ISP2** to solve this problem heuristically. The time required is $O(n \log n)$, as before.

Theorem 13. Let b^H be the heuristic value of the solution to an instance of $B|t_{ij} = U_j|\max_{1 \leq j \leq n}\{b_j\}$ obtained by Algorithm **ISP2**, and let b^* be the optimal value for that instance. Then, $b^H/b^* \leq 2$, and this bound is asymptotically attainable.

Proof. A rather lengthy proof is given in [11]. We outline this proof below:

For any solution (optimal or otherwise) $\Pi = (\mathcal{I}, \gamma)$ for any broadcast problem, we denote by $(b_1, \dots, b_n)$ the vector of reception times. Let $\Pi^* = (\mathcal{I}^*, \gamma^*)$ denote the solution obtained by ISP2, and let b_j^* be the corresponding reception times.

1. There are $\lceil n/2 \rceil$ leaves in the solution Π^* and these are precisely the vertices in the leaf set $\Lambda = \{j | n+1 - \lceil n/2 \rceil \leq j \leq n\}$. Also, $j < k$ implies that $b_j^* \leq b_k^*$.
2. Let OPT = the optimal value of the objective function for problem $B|t_{ij} = U_j|\max_{1 \leq j \leq n}\{b_j\}$. Then, $LB = \sum_{j \in \Lambda} b_j^* \lceil n/2 \rceil$ is a lower bound on OPT . U_n is also a lower bound on OPT . Hence, $\max\{LB, U_n\}$ is a lower bound on OPT .
3. Since LB is the mean of the values $b_{n+1-\lceil n/2 \rceil}^*, \dots, b_n^*$, it follows that $b_{n+1-\lceil n/2 \rceil}^* \leq LB \leq b_n^*$.
4. From Step 2 of ISP2, $b_n^* - b_j^* \leq U_n$ for $j \in \Lambda$.
5. Hence, $\max_{1 \leq j \leq n}\{b_j^*\} = b_n^* \leq b_{n+1-\lceil n/2 \rceil}^* + U_n \leq LB + U_n$. Also, $\text{OPT} \geq \max\{LB, U_n\}$. Thus, $\max_{1 \leq j \leq n}\{b_j^*\}/\text{OPT} \leq (LB + U_n)/\max\{LB, U_n\} \leq LB/LB + U_n/U_n = 2$.
6. Tightness of the bound is proved by an example with $n = 2^h$, $U_i = 1$ for $1 \leq i \leq 2^h - 1$ and $U_n = h$. It is easy to show that $\text{OPT} = h$ while ISP2 yields a value of $2h - 2$. Asymptotically, the ratio $(2h - 2)/h \rightarrow 2$ as $h \rightarrow \infty$. ■

5.2. $B|t_{ij} = U_j|b^t$

The computational complexity of this problem remains open. We propose a heuristic procedure GREEDY and provide an upper bound on its worst-case performance. GREEDY adds a vertex i to the solution by choosing as the direct predecessor of i a vertex k already in the solution so that this assignment will either not lead to an increase in b^t or will cause the minimal possible increase in b^t . After the i th iteration of Step 1, c_j is the cycle time for vertex j in the current partial solution tree, and $\max\{c_j | 1 \leq j \leq i\}$ is the corresponding value of b^t .

Heuristic GREEDY

Given: positive numbers U_j , $0 \leq j \leq n$, with $U_1 \leq \dots \leq U_n$.

0. Initialization:

Starting broadcast tree defined by $T = \{0\}$ and $A(T) = \emptyset$.

Set $c_j = 0$ for $0 \leq j \leq n$.

1. For $i = 1, \dots, n$

Let $\alpha = \min\{c_j + U_j | j \in T\}$.

Let k be the smallest index which yields α . Assign k to transmit next to i .

Let $c_k = \alpha$ and let $c_i = U_k$.

$T = \{0, \dots, i\}$.

$A(T) = A(T) \cup \{(k, i)\}$, $\gamma_k = (\gamma_k, i)$.

EndFor.

Terminate.

Using a heap to store the data for the current solution, the above algorithm can be implemented in $O(n \log n)$ steps. To analyze the worst-case performance of GREEDY, we prove the following preliminary result:

Lemma 8. *There exists an optimal solution to problem $B|t_{ij} = U_i|b^t$ in which the leaf vertices may be characterized as $\{j | t < j \leq n\}$ for some $t \in \{1, \dots, n\}$.*

Proof. Recall that $U_1 \leq \dots \leq U_n$. Let $\Pi = (\mathcal{T}, \gamma)$ be any (not necessarily optimal) solution (i.e., any broadcast tree for the problem). Assume that in Π we have indices h and k with $h < k$ and such that h is a leaf and k is not a leaf of $\mathcal{T}$. Let $\Pi' = (\mathcal{T}', \gamma')$ be obtained from Π by switching the positions of h and k . We shall show that $b^{t'} \leq b^t$, that is, the cycle time for Π' is at least as small as that of Π . Repeated application of this process establishes the lemma.

CASE 1. $U_h = U_k$.

In this case, it is obvious that $b^{t'} \leq b^t$.

CASE 2. $U_h < U_k$.

For any vertex $e \in T$, let $p(e)$ be the vertex which transmits to e (for vertex 0, $p(e)$ is not defined), and, as usual, let $s(e)$ be the set of vertices to which e transmits. Then, the busy time for e is $c_e = U_{p(e)} + |s(e)|U_e$, where $U_{p(0)} \equiv 0$. Thus, e contributes $|s(e)|U_e$ to the calculation of c_e and e also contributes U_e to the calculation of c_m for $m \in s(e)$. Hence, a decrease in U_e for e a leaf vertex will not change any of the c_j 's since $|s(e)| = 0$ for a leaf, while a decrease in c_e for e not a leaf vertex will decrease the c_j 's for $j \in \{e\} \cup s(e)$. No other c_j 's change.

Now without changing $\Pi = (\mathcal{T}, \gamma)$, we will change the values of U_k and U_h . Let $\Delta = U_k - U_h > 0$. We decrease U_k by Δ (which has the effect of reducing U_k to the original value of U_h) and increase U_h by Δ (which has the effect of increasing U_h to the original value of U_k). By the argument of the previous paragraph, there will be decreases in c_j for $j \in k \cup s(k)$, and all other c_j 's remain unchanged. It follows that the cycle time b^t for the problem defined by the new set of U_j 's is no greater than the original cycle time b^t . Similarly, we may exchange the positions of h and k in solution Π without increasing the cycle time, thus yielding $\Pi' = (\mathcal{T}', \gamma')$. This completes the proof. ■

Theorem 14. *Let b^H be the heuristic value of the solution to an instance of problem $B|t_{ij} = U_i|b^t$ obtained by Heuristic GREEDY, and let b^* be the optimal value for that instance. Then, $b^H/b^* \leq 2$, and this bound is asymptotically attainable.*

Proof. Let $\Pi = (\mathcal{T}, \gamma)$ be an optimal solution which satisfies Lemma 8, and based upon Lemma 8, let $\{t + 1, \dots, n\}$ be the leaf vertices of $\mathcal{T}$. Note that the sets $s(j)$ of direct successors of j in Π satisfy

$$\sum_{0 \leq j \leq t} \sigma(j) = n, \quad (4)$$

since $\sigma(j) = 0$ for $j \geq t + 1$. Since $\sigma(j) > 0$ for $0 \leq j \leq t$, it follows that

$$U_j \leq b^*, \quad 0 \leq j \leq t. \quad (5)$$

It suffices to show that GREEDY produces a solution with cycle time $\leq 2b^*$.

We claim that, at the end of the first t repetitions of Step 1, the cycle time for the partial solution is $\leq 2b^*$. Let $j \leq t$. To verify the claim, it suffices to show that if Step 1 assigns k to transmit to j then at the end of this transmission c_k and c_j are both $\leq 2b^*$. Since in Step 1 j is assigned to receive its transmission from the vertex which minimizes the resulting cycle time, it suffices to show that there is at least one vertex $h \in \{0, \dots, j$

$-1\}$ which can transmit to j with the result that c_h and c_j are both $\leq 2b^*$. We may choose h to be any leaf vertex of the tree on $\{0, \dots, j-1\}$, since for such a vertex c_h (after the transmission) will equal $U_{p(h)} + U_h$, which by (5) is $\leq 2b^*$, and c_j will equal $U_j \leq b^*$, also by (5). Thus, at the end of the first t repetitions of Step 1, the cycle time is $\leq 2b^*$.

We now examine the assignment of the set $\{t+1, \dots, n\}$. Observe that the number of direct successors that a vertex j can have without increasing the cycle time beyond $2b^*$ is precisely $\lfloor (2b^* - U_{p(j)})/U_j \rfloor$. This is because vertex j is occupied for $U_{p(j)}$ units of time receiving the message and each outgoing message occupies vertex j for U_j time units. Thus, at the start of the $(t+1)$ st repetition of Step 1 of GREEDY, there are

$$\sum_{j=0}^t \lfloor (2b^* - U_{p(j)})/U_j \rfloor - t \quad (6)$$

available positions for additional vertices to be added to the current tree without exceeding a cycle time of $2b^*$, where the term $-t$ at the end of (6) accounts for the already assigned vertices $\{1, \dots, t\}$. Application of (5) as shown below to expression (6) provides a bound on (6) for $i = t+1$ (note that since all j 's in the sum below are $\leq t$, it follows that all $U_{p(j)}$'s are $\leq b^*$): $\sum_{j=0}^t \lfloor (2b^* - U_{p(j)})/U_j \rfloor - t \geq \sum_{j=0}^t \lfloor b^*/U_j \rfloor - t \geq \sum_{j=0}^t \sigma(j) - t = n - t$.

The last inequality above holds since $\sigma(j)$, the number of direct successors of j in the optimal solution $\Pi = (\mathcal{T}, \gamma)$, must satisfy $\sigma(j) \leq \lfloor b^*/U_j \rfloor$ in order for b^* to be the optimal value of b^t . The last equality above is relation (4). Thus, at the completion of the t th iteration, there are at least $n - t$ available positions to place the remaining $n - t$ vertices $\{t+1, \dots, n\}$ without violating the bound of $2b^*$. Since Step 1 will choose the assignment which minimizes cycle time at each iteration, it follows that the vertices $t+1, \dots, n$ will be assigned in a manner which will not lead to a violation of the $2b^*$ bound. This completes the proof of the upper bound in the theorem.

To see that the bound is attainable, consider the following instance of $B|t_{ij} = U_i|b^t$: $n = 2M + 2$, $U_0 = M$, $U_1 = 1$, $U_2 = \sqrt{M}$, $U_3 = U_4 = \dots = U_{2M+2} = M$, where $M \geq 1$. The reader can verify that in one optimal solution vertex 0 transmits to 2, 2 transmits to 1, 1 transmits to M of the other vertices (say, 3 to $M+2$), and each of these vertices transmits to one of the M remaining vertices (say, $M+3$ to $2M+2$). The resulting optimal cycle time is $M + \sqrt{M}$, which is the value of both c_1 and c_2 .

When GREEDY is applied to this instance, it will start by assigning 0 to transmit to 1 and 1 to transmit to 2. We shall calculate a lower bound LB on the cycle time of all feasible solutions which have transmissions from 0 to 1 and from 1 to 2. Obviously, LB will apply to the

solution obtained by GREEDY. From now on we assume that 0 transmits to 1 and 1 transmits to 2 in some arbitrary feasible solution $\Pi' = (\mathcal{T}', \gamma')$. Recalling that we do not include transitive arcs in $A(T')$, we define $x = |\{(0, j) \in A(T') | 3 \leq j \leq 2M + 2\}|$, $y = |\{(1, j) \in A(T') | 3 \leq j \leq 2M + 2\}|$, and $z = |\{(2, j) \in A(T') | 3 \leq j \leq 2M + 2\}|$. We consider three cases:

CASE 1. $x \geq 1$.

In this case, vertex 0 makes at least two transmissions, so $c_0 \geq 2U_0 = 2M$, and, hence, $b^t \geq 2M \equiv LB1$.

CASE 2. $x = 0$ and $y + z < M$.

Without loss of generality, we may assume that $Q = \{3, \dots, 2 + x + y\}$ is the set of vertices in $\{3, \dots, 2M + 2\}$ which receives direct transmissions from 1 or 2. The $2M - (y + z)$ remaining vertices in the set $R = \{3 + y + z, \dots, 2M + 2\}$ must each have a predecessor (not necessarily direct) in the set Q . Since $|R| > M > |Q|$, there must be some $i \in Q$ which has at least 2 successors. Hence, one of the following situations must exist:

- (i) There exist $j, k \in R$ with $(i, j) \in A(T')$ and $(j, k) \in A(T')$. In this case, $c_i \geq U_i + U_j = 2M$.
- (ii) There exist $j, k \in R$ with $(i, j) \in A(T')$ and $(i, k) \in A(T')$. In this case, $c_i \geq 2U_i = 2M$.

In either situation (i) or (ii) above, $b^t \geq 2M \equiv LB2$.

CASE 3. $x = 0$ and $y + z \geq M$.

If $y \geq M[(\sqrt{M} - 1)/(\sqrt{M} + 1)]$, then $c_1 \geq M + 1 + y \geq M + 1 + M[(\sqrt{M} - 1)/(\sqrt{M} + 1)]$. Alternatively, if $y < M[(\sqrt{M} - 1)/(\sqrt{M} + 1)]$, then $z \geq M - y > M - M[(\sqrt{M} - 1)/(\sqrt{M} + 1)]$ and $c_2 \geq 1 + z\sqrt{M} > 1 + (M - M[(\sqrt{M} - 1)/(\sqrt{M} + 1)])\sqrt{M} = M + 1 + M[(\sqrt{M} - 1)/(\sqrt{M} + 1)]$.

Thus, regardless of what the value of y is in Case 3, we get $b^H \geq M + 1 + M[(\sqrt{M} - 1)/(\sqrt{M} + 1)] \equiv LB3$.

Since one of the three cases above must hold for the GREEDY solution, we get

$$\begin{aligned} b^H &\geq \min\{LB1, LB2, LB3\} \\ &= \min\{2M, 2M, M + 1 \\ &\quad + M[(\sqrt{M} - 1)/(\sqrt{M} + 1)]\} \\ &= M + 1 + M[(\sqrt{M} - 1)/(\sqrt{M} + 1)]. \end{aligned}$$

Thus, $b^H/b^* \geq (M + 1 + M[(\sqrt{M} - 1)/(\sqrt{M} + 1)])/(M + \sqrt{M})$, which implies that $\lim_{M \rightarrow \infty} b^H/b^* = 2$. ■

6. SUMMARY AND CONCLUSIONS

This work contains a number of results for broadcasting with nonunit transmission times with a fixed vertex as the broadcaster. It is interesting to note that, in their review paper discussed in Section 1, Hedetniemi et al. [5] identified a number of areas for further study in gossiping and broadcasting theory, and among these is the generalization of the unit transmission time assumption. This assumption is used in the great majority of the related published work.

Two of our objective functions are classical (the maximum broadcast time and the cycle time). The other three reflect the realities of many active systems, and we hope that our initial efforts will encourage others to expand their ranges of study.

The majority of problems studied here are proved to be computationally difficult; thus, heuristic approaches, as illustrated in Section 5, represent an important area for further investigations. Other transmission time function models may also be worth investigating, for example, the additive model with $t_{ij} = U_i + U_j$ and the multiplicative model with $t_{ij} = U_i \cdot U_j$. Since it is likely that almost all resulting models will prove to be computationally intractable, the research focus would again be on heuristics.

More generally, other problems have been studied under the unit transmission time assumption. Generalization of these results would be worthwhile. For example, there is great interest in systems in which all vertices may be broadcasters. For such problems, many objectives have been proposed, for example, minimization of the number of arcs needed to guarantee that the broadcast time from each vertex is no greater than a fixed value.

The gossiping problem is similar to the broadcasting problem and deals with situations in which all members of a set must transmit a unique piece of information to all other members. This is a natural direction in which our work could be extended.

Thus, the field for further development of this research is wide and promising, and the authors encourage others to join in the effort.

APPENDIX

Proof of Theorem 6. By reduction from the following problem known to be unary NP-complete. Numerical 3-Dimensional Matching [3, p. 224]: Given three disjoint sets $X = \{x_1, \dots, x_m\}$, $Y = \{y_1, \dots, y_m\}$, and $Z = \{z_1, \dots, z_m\}$ and an integer $b = (\sum_{j \in X} x_j + \sum_{j \in Y} y_j + \sum_{j \in Z} z_j)/m$, can $X \cup Y \cup Z$ be partitioned into m disjoint sets $A_1, \dots, A_m$, such that A_i contains exactly one element from each of X , Y , and Z and such that the sum of the elements in A_i equals b , $i = 1, \dots, m$?

Define $\bar{x} = \sum_{j \in X} x_j/m$, $\bar{y} = \sum_{j \in Y} y_j/m$, $\bar{z} = \sum_{j \in Z} z_j/m$,

$y_{\max} = \max_{1 \leq j \leq m} \{y_j\}$, and $z_{\max} = \max_{1 \leq j \leq m} \{z_j\}$. Consider the following instance of $B|t_{ij} = U_i, r_j|\sum_{j=1}^n b_j$:

$$n = 6m - 1$$

We have the following vertices:

- (R_0) 1 vertex (0), with $r_0 = 0$, $U_0 = M^4$
- (R_1) $m - 1$ vertices with $r_j = 0$, $U_j = M^4$
- (R_2) m vertices with $r_j = kM^4$, $U_j = M^3 + x_j$
- (R_3) m vertices with $r_j = (k + 1)M^4$, $U_j = M^2 + y_j$
- (R_4) m vertices with $r_j = (k + 1)M^4 + M^3$, $U_j = M + z_j$
- (R_5) m vertices with $r_j = (k + 1)M^4 + M^3 + M^2$, $U_j = M$
- (R_6) m vertices with $r_j = (k + 1)M^4 + M^3 + M^2 + M + b$, $U_j = M^5$

$$\begin{aligned} C = & [M^4 \sum_{j=1}^k j 2^{j-1}] + [m(k + 1)M^4] \\ & + \sum_{j=1}^m [(k + 1)M^4 + M^3 + x_j] \\ & + \sum_{j=1}^m [(k + 1)M^4 + M^3 + M^2 + x_j + y_j] \\ & + m[(k + 1)M^4 + M^3 + M^2 + M + b] \\ & + m[(k + 1)M^4 + M^3 + M^2 + 2M + b]. \end{aligned}$$

Proof of Theorem 7. By reduction from Numerical 3-Dimensional Matching (as defined in the proof of Theorem 6). Instance of $B|t_{ij} = U_j, r_j|\sum_{j=1}^n b_j$:

$$n = 16m + 3$$

- (R_0) 1 vertex (0) with $r_0 = 0$, $U_0 = 1$
- (R_1) m vertices with $r_j = j - 1$, $U_j = 1$
- (R_2) 1 vertex with $r_j = m$, $U_j = M^4 - m + b$
- (R_3) m vertices with $r_j = j$, $U_j = M - j$
- (R_4) m vertices with $r_j = M$, $U_j = M^4 - M + b$
- (R_5) m vertices with $r_j = M$, $U_j = M^2 + x_j$
- (R_6) m vertices with $r_j = M^2 + M + x_j$, $U_j = M^4 - (M^2 + M + x_j) + b$
- (R_7) m vertices with $r_j = M^2 + M$, $U_j = M^3 + y_j$
- (R_8) $2m$ vertices with $r_j = M^3 + M^2 + M$, $U_j = M^4 - (M^3 + M^2 + M) + z_j$
- (R_9) $8m + 2$ vertices with $r_j = M^4 + b$, $U_j = M^5$

$$\begin{aligned} C = & [m(m + 1)/2] + 4m[M^4 + b] + [mM] \\ & + [m(M^2 + M) + \sum_{j=1}^m x_j] \\ & + [m(M^3 + M^2 + M) + \sum_{j=1}^m (x_j + y_j)] \\ & + [M^4 + b] + [(8m + 2)(M^5 + M^4 + b)]. \end{aligned}$$

Proof of Theorem 9. By reduction from Numerical 3-Dimensional Matching (as defined in the proof of Theorem 6). Instance of $B|t_{ij} = U_j|\sum_{j=1}^n w_j b_j$:

$$n = 8m - 1$$

We have the following vertices:

- (R_0) 1 vertex (0), where U_0 and w_0 are arbitrary
- (R_1) $m - 1$ vertices with $U_j = M$, $w_j = M^2$
- (R_2) m vertices with $U_j = M + x_j$, $w_j = 4M + x_j$
- (R_3) $2m$ vertices with $U_j = 2M + y_j$, $w_j = 2M + y_j/2$
- (R_4) $4m$ vertices with $U_j = 4M + z_j$, $w_j = M + z_j/4$

$$C = M^3 \sum_{j=1}^k j 2^{j-1} + M^2 m [4(k+1) + 4(k+3) + 4(k+7)] + Mm[(k+13)\bar{x} + (k+11)\bar{y} + (k+11)\bar{z}] + \sum_{j=1}^m (x_j^2 + y_j^2 + z_j^2)/2 + mb^2/2.$$

Proof of Theorem 10. By reduction from Numerical 3-Dimensional Matching (as defined in the proof of Theorem 6). Instance of $B|t_{ij} = U_i|\max_{1 \leq j \leq n} \{b_j\}$:

$$n = 16m - 1$$

We have the following vertices:

- (R_0) m vertices, including vertex 0, with $U_j = M^2$
 - (R_1) m vertices with $U_j = M^2 + M + x_j$
 - (R_2^a) m vertices with $U_j = M^2 + 2M + y_j$
 - (R_2^b) m vertices with $U_j = M^2 + 2M + y_{\max}$
 - (R_3^a) m vertices with $U_j = M^2 + 4M + z_j$
 - (R_3^b) $3m$ vertices with $U_j = M^2 + 4M + z_{\max}$
 - (R_4) $8m$ vertices with $U_j = C + 1$
- $$C = (k+4)M^2 + 7M + b.$$

Proof of Theorem 11. By reduction from Numerical 3-Dimensional Matching (as defined in the proof of Theorem 6). Instance of $B|t_{ij} = U_j|\max_{1 \leq j \leq n} \{b_j\}$:

$$n = 8m - 1$$

We have the following vertices:

- 1 vertex (0) with U_0 arbitrary

- $m - 1$ vertices with $U_j = M^2$
- m vertices with $U_j = M^2 + M + x_j$
- $2m$ vertices with $U_j = M^2 + 2M + y_j$
- $4m$ vertices with $U_j = M^2 + 4M + z_j$
- $C = (k+3)M^2 + 7M + b.$

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